Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"
Service Contract N° 945120-IPR-2023 (JRC)

VALIDATION REPORT
UPDATE 2024

SOIL WATER INDEX
VERSION 1.0

Issue I1.00

Organization name of lead contractor for this deliverable: TU Wien

Book Captain: Bernhard Raml
Contributing Authors: Samuel Massart
Shahrzad Afroozeh
Bernhard Bauer-Marschallinger

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

© C-GLOPS Lot1 consortium

Document Release Sheet

Book captain:	Bernhard Bauer-Marschallinger (TU Wien)	Sign Bernhard Bauer-Marschallinger Date 16.05.2025
Approval:	Roselyne Lacaze (HYGEOS)	Sign Roselyne Lacaze Date 05.06.2025
Endorsement:	Nadine Gobron (JRC)	Sign Date
Distribution:	Public	

Change Record

Issue/Rev	Date	Page(s)	Description of Change	Release
	16.05.2025	All	First issue	I1.00

© C-GLOPS Lot1 consortium

TABLE OF CONTENTS

Executive Summary **14**

1 Background of the document **16**

1.1 Scope and Objectives **16**

1.2 Content of the document **16**

1.3 Related documents **16**

1.3.1 Applicable documents 16
1.3.2 Input 17
1.3.3 Output 17
1.3.4 External documents 17

2 Users Requirements **18**

2.1 specific technical details and requirements **18**

2.2 Further requirements: **19**

2.2.1 Output product composition 19
2.2.2 Data Structure 19
2.2.3 Data format 19
2.2.4 Uncertainties and Validation 20
2.2.5 Input data 20
2.2.6 Product delivery 21

3 Review of initial SWI1km Validations **22**

4 Scientific Quality Monitoring Method **24**

4.1 In-situ Reference Products **24**

4.1.1 Used ISMN Networks 25
4.1.2 Soil Depth Levels 28
4.1.3 Spatial Representativeness 32

4.2 Validation procedure **33**

4.2.1 Dataset Preparations 33
4.2.2 Assessment of the Temporal Signal Quality 34

5 Results **36**

5.1 Overall Analysis Summary **36**

5.1.1 Summary over Networks: Pearson Correlation Analyses 38

5.1.2 Summary over Networks: RMSD Analyses 47
5.2 Selected Time Series Analyses 51
5.2.1 Near-Surface Soil Layer (T-Value = 2) 51
5.2.2 Sub-Surface Soil Layer (T-Value = 10) 55
5.2.3 Deep Soil Layer (T-Value = 100) 58
6 Discussions and Conclusions 62
6.1 Summary from Dedicated Peer-Review Article 62
6.2 Quality Assessment over Europe 2015-2024 62
7 Recommendations and Perspectives 64
8 References 65

© C-GLOPS Lot1 consortium

List of Figures

Figure 1: Locations of input ISMN stations (colored dots) for the 2024's SWI1km analysis available at the near-surface. 29

Figure 2: Locations of input ISMN stations (colored dots) for the 2024's SWI1km analysis available at the sub-surface. 30

Figure 3: Locations of input ISMN stations (colored dots) for the 2024's SWI1km analysis available at the deep soil. 31

Figure 4: Boxplot summarizing Pearson rho values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) for the period between 2015 and 2024, per network. Median values are represented by the horizontal line. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points. 40

Figure 5: Box plot summarizing Pearson rho values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) per network, based on data from 2015-2023 (brown) and 2024 (orange). Only networks with data covering 2024 are included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points. 41

Figure 6: Pearson correlation (rho) for SWI1km with T-value=2 over selected ISMN networks (0-10 cm depth). See the land cover classes in Table 4. 42

Figure 7: Boxplot summarizing Pearson rho values calculated for SWI1km with T-value=10 and all relevant ISMN stations (10-30 cm depth) for the period between 2015 and 2024, per network. 43

Figure 8: Box plot summarizing Pearson rho values calculated for SWI1km with T-value=10, and all suitable ISMN stations (10-30 cm depth) per network, based on data from 2015-2023 (brown) and 2024 (orange). Only networks with data covering 2024 are included. 43

Figure 9: Pearson correlation (rho) for SWI1km with T-value=10 over selected ISMN networks (10-30 cm depth). See the land cover classes in Table 4. 44

Figure 10: Boxplot summarizing Pearson rho values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) for the period between 2015 and 2024, per network. 45

Figure 11: Box plot summarizing Pearson rho values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) per network, based on data from 2015-2023 (brown) and 2024 (red). Only networks with data covering 2024 are included. 45

Figure 12: Pearson correlation (rho) for SWI1km with T-value=100 over selected ISMN networks (30-80 cm depth). See the land cover classes in Table 4. 46

Figure 13: Boxplot summarizing RMSD values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) for the period between 2015 and 2024, per network. 48

Figure 14: Box plot summarizing RMSD values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) per network, based on data from 2015-2023 (dark green) and 2024 (light green). Only networks with data covering 2024 are included. 48

Figure 15: Boxplot summarizing RMSD values calculated for SWI1km with T-value=10 and all relevant ISMN stations (10-30 cm depth) for the period between 2015 and 2024, per network. 49

Figure 16: Box plot summarizing RMSD values calculated for SWI1km with T-value=10, and all suitable ISMN stations (10-30 cm depth) per network, based on data from 2015-2023 (dark green) and 2024 (light green). Only networks with data covering 2024 are included. 49

Figure 17: Boxplot summarizing RMSD values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) for the period between 2015 and 2024, per network. 50

Figure 18: Box plot summarizing RMSD values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) per network, based on data from 2015-2023 (dark blue) and 2024 (light blue). Only networks with data covering 2024 are included. 50

Figure 20: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at FMI station SOD012 at depth = 0.10m (tree cover – needle leaved). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left. 52

Figure 21: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at TERENO station Gevenich at depth = 0.05m (cropland - rainfed) 52

Figure 22: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at TERENO station Schoeneseiffen at depth = 0.20m (grassland). 53

Figure 23: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at TERENO station Wildernath at depth = 0.05m (Forest – broad leaved). 53

Figure 24: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at XMS-CAT station Serra De Costa Ampla at depth = 0.05m (cropland - rainfed). 53

Figure 25: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at REMEDHUS station Carretoro at depth = [0-0.05]m (cropland - rainfed). 54

Figure 26: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at REMEDHUS station La Cruz De Elias at depth = [0-0.05]m (cropland - rainfed)..... 54

Figure 27: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at RSMN station Iasi at depth = [0-0.05]m (cropland - rainfed)..... 54

Figure 28: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at RMSN station Satu Mare at depth = [0-0.05]m (cropland - rainfed)..... 55

Figure 29: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at COSMOS-UK station Balruderry at depth = 0.10m (cropland - rainfed). 55

Figure 30: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at STEM station CAN-GC-T at depth = 0.30m (cropland - rainfed). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left. 56

Figure 31: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at FMI station SOD101 at depth = 0.20m (tree cover – needle leaved). 57

Figure 32: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at TERENO station Wildenrath at depth = 0.20m (Cropland - rainfed). 57

Figure 33: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at TERENO station Selhausen at depth = 0.20m (cropland - rainfed). 57

Figure 34: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at XMS-CAT station SerraDeCostaAmpla at depth = 0.20m (cropland - rainfed)..... 58

Figure 35: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at NVE station As at depth = 0.50m (cropland - rainfed). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left..... 59

Figure 36: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at STEMS station CAN-GC-T at depth = 0.30m (cropland - rainfed)..... 60

Figure 37: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at XMS-CAT station Serra de Costa Ampla at depth = 0.50m (cropland - rainfed). 60

Figure 38: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at XMS-CAT station Batlliu de Sort at depth = 0.50m (grassland)..... 60

Figure 39: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at TERENO station Schoeneseiffen at depth = 0.70m (grassland). 61

Figure 40: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at TERENO station Gevenich at depth = 0.30m (cropland). 61

Figure 41: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at FMI station SOD130 at depth = 0.40m (cropland). 61

© C-GLOPS Lot1 consortium

List of Tables

Table 1: ISMN networks used in the present SWI1km validation campaign with characteristics as given in the ISMN documentation. At some stations, multiple sensors are installed at the same depth. Each of these sensors is included in the number of suitable stations. 26

Table 2: Depth levels for validation experiments. 28

Table 3: Overall summary of statistics from the time series analysis of SWI1km versus in-situ measurements: Mean values of Pearson rho and RMSD, per network, and total per soil layer. The number of included in-situ stations is given by n. 37

Table 4: Background land cover types for Figure 6 to Figure 12, derived from the ESA WorldCover map 2020. 41

© C-GLOPS Lot1 consortium

List of Acronyms

AD	: Applicable document
API	: Application Programming Interface
ASCAT	: Advanced Scatterometer
ATBD	: Algorithmic Theoretical Basis Document
BfG	: German Federal Institute of Hydrology
BIEBRZA	: In-Situ Network of the Instytut Geodezji i Kartografii, Poland.
CAL/VAL	: Calibration and Validation
CEOS	: Committee on Earth Observation Satellites
CGLOPS	: Copernicus Global Land Operations
CGLS	: Copernicus Global Land Service
COG	: Cloud Optimized Geotiff
D03	: IMSN flag GLDAS soil temperature < 0°C
D05	: IMSN flag soil moisture shows peaks without precipitation event (GLDAS) in the preceding 24 hours
D10	: IMSN flag saturated plateau (for a minimum time length of 12 hours) occurs in soil moisture spectrum
DWD	: Deutscher Wetterdienst
CSAR	: C-band Synthetic Aperture Radar
ECMWF	: European Centre for Medium-Range Weather Forecasts
ERA5	: European Reanalysis version 5
EU	: European Union
FMI	: Finnish Meteorological Institute
FR-Aqui	: France Institute of Agricultural Research
GBOV	: Ground-Based Observations for Validation
GCOS	: Global Climate Observing System
GLDAS	: Global Land Data Assimilation System
GROW	: In-Situ Network of the GROW Observation Network
HOBE	: In-Situ Network of the Center for Hydrology – Hydrological Observatory
ICWRGC	: International Center for Water Resources and Global Change
ISMN	: International Soil Moisture Network
IMA	: Institute for Agricultural and Earthmoving Machines
INSPIRE	: Infrastructure for spatial information in Europe
IPE	: Instituto Pirenaico de Ecologia
JRC	: Joint Research Centre of the European Union
LPV	: Land Product Validation
Metop	: Meteorological Operational Satellite
MOL-RAO	: Meteorologische Observatorium Lindenberg - Richard-Aßmann-Observatorium
NETCDF	: Network Common Data Form

Noah-LSM	: Noah Land Surface Model
NRT	: Near Real Time
PQAD	: Product Quality Assurance Document
PUM	: Product User Manual
QAR	: Quality Assessment Report
rho	: Pearson correlation coefficient (rho)
RHEMEDUS	: In-Situ Network of the Centro Hispano Luso de Investigaciones Agrarias, Salamanca, Spain
RMSD	: Root Mean Square Difference
RSMN	: In-Situ Network of the National Meteorological Administration, Romania
SAR	: Synthetic Aperture Radar
SCATSAR	: Scatterometer - Synthetic Aperture Radar
SQAD	: Service Quality Assurance Document
SM	: Soil Moisture
SM2RAIN	: Soil Moisture to Rain
SQE	: Scientific Quality Evaluation
SSM	: Surface Soil Moisture
STEMS	: In-Situ Network of the Institute of Sciences and Technologies for Sustainable Energy and Mobility
SWBM	: Soil Water Balance Model
SWI	: Soil Water Index
TERENO	: In-Situ Network of the Terrestrial Environmental Observatories
UNFCCC	: United Nations Framework Convention on Climate Change
VR	: Validation Report
WEGENERNET	: In-Situ Network of the Wegener Center for Climate and Global Change, Graz, Austria
WGS	: World Geodetic System
WMO	: World Meteorological Organisation
WSMN	: Wales Soil Moisture Network

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. The production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This document describes the Scientific Quality Monitoring of the 1km Soil Water Index (SWI1km) product for the year 2024. In accordance with the System Quality Assurance Document (SQAD) [CGLOPS1_SQAD], the soil moisture quality monitoring within the CLMS has been transformed and further automated, the product quality analysis results are compiled as an annual update of the Quality Assessment Report (QAR).

The aim of this annual analysis is to assess the performance of the near real time (NRT) SWI1km product in the period from 2024-01-01 until 2024-12-31 relative to the performance of the SWI1km archive available from the CLMS of the previous years, starting in 2015. The validation presented in this report was performed according to the procedures defined in the Product Quality Assurance Document [CGLOPS1_PQAD].

Validation and continuous Quality Monitoring constitute the only means of guaranteeing the compliance of generated products with user requirements: the former concerns the new products which must pass an exhaustive scientific evaluation before operational implementation. The latter concerns with keeping the quality of operational products at the same level along the time. Both follow, as much as possible, the guidelines, protocols and metrics defined by the Land Product Validation (LPV) group of the Committee on Earth Observation Satellite (CEOS) for the validation of satellite-derived land products.

In this report, the SWI1km product, based on Scatterometer-Synthetic Aperture Radar Soil Water Index (SCATSAR-SWI) algorithm that ingests Metop ASCAT and Sentinel-1 Surface Soil Moisture (SSM), is subject to quality assessment experiments. For the period 2015-2023 and the year 2024, the dataset is compared to in-situ data from suitable networks over Europe of the International Soil Moisture Network (ISMN). The analyses comprise examinations of the temporal dynamics of the SWI1km and how it agrees with the ground reference data. Taking into account the scale gap between in-situ point measurement and the areal satellite data and expected spatial representativeness uncertainty, the obtained results are overall satisfying, with scores comparable with the ASCAT-alone based product SWI V3 (see CGLOPS1_VR2024_SWIV3), and much improved scores against the

Sentinel-1-alone based product (see CGLOPS1_VR2024_SSM1km). The comparison of data from the year 2024 versus data from 2015-2023 shows a quality consistency for all networks and soil layers, also confirming what can be expected from stable quality of the Sentinel-1 input data in the recent years.

The SWI1km demonstrates a very good ability to reproduce peaks and lows of the ground soil moisture at the near-surface layer, and the data fusion approach seem to be most beneficial for capturing temporal dynamics in the 24h-time-scale. While capturing the general temporal signal, this does not hold true for the deeper layers, with slightly less accurate reproduction of peaks and lows, and partially with time lags (delays) between satellite and in-situ time series.

Inheriting the parametrisation of the input SSM from ASCAT and Sentinel-1, the SWI1km shows seasonal biases against some ISMN stations, related to vegetation dynamics. Also, in some areas with cold climates, the agreement with ground measurements was found to be rather low. This is most likely caused by the evergreen needle leaf forest and partly inundated land cover dominating the region. However, the issue still needs further investigation, focusing on the precision of the frozen/snow-masking, and an examination of the reliability of the local ground measurements and their flagging.

The loss of Sentinel-1B in late 2021 has been acknowledged, while Sentinel-1A is stably observing, and Sentinel-1C was successfully launched at the end of 2024, and is expected to be fully operational by mid of 2025. In the data for 2022 to 2024, it could be confirmed that the SWI-approach is able to cope with the inconsistencies in the spatiotemporal coverage of its SAR component. The SWI1km signal quality is only weakly affected, with no degradation of the scores.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document presents the results of the annual scientific quality monitoring of Soil Water Index 1km Version 1 product (named hereafter SWI1km) derived from Sentinel-1 CSAR and Metop ASCAT data. This product provides daily soil moisture images over Europe from January 2015 ongoing. The quality evaluation is performed on SWI1km data from 2024-01-01 to 2024-12-31 with a reference period from 2015-01-01 to 2023-12-31.

The objective is to check that the operational products keep the same level of quality in the period under study as the fully validated products.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements, and the expected performance
- Chapter 3 reviews the results of previous validation reports
- Chapter 4 describes the methodology for quality assessment, the metrics and the criteria of evaluation
- Chapter 5 presents the results of the analysis
- Chapter 6 summarizes the main conclusions of the study
- Chapter 7 makes recommendations based upon the results

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for global Vegetation and Energy products of the Copernicus Land Monitoring Service.
CGLOPS1_SQAD	System Quality Assurance Document for global Vegetation and Energy products of the Copernicus Land Monitoring Service.
CGLOPS1_ATBD_SWI1km-V1	Algorithm Theoretical Basis Document of the Soil Water Index 1kmVersion 1 product.
CGLOPS1_QAR_SWI1km-V1	Quality Assessment Report for the Soil Water Index 1km Version 1 product.
CGLOPS1_VR2024_SWIV3	Validation Report of the Soil Water Index 0.1° Version 3 product for 2024.
CGLOPS1_VR2024_SSM1km	Validation Report of the Surface Soil Moisture 1km for 2024.

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_ SWI1km-V1	Product User Manual summarizing all information about the Soil Water Index 1km V1 product.

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS-245	The 2022 GCOS ECVs Requirements

Available at: https://library.wmo.int/idurl/4/58111

2 USERS REQUIREMENTS

According to the applicable documents [AD1], the requirements relevant for Soil Water Index 1km are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

SWI – Soil Water Index	
The surface soil water index provides daily information about moisture conditions in different soil depths. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation. The SWI quantifies the amount of water (m^3/m^3) in soil layers at various depths and is derived from the SSM using a water balance model. The freeze and thaw status of soil is provided.	
Geometric Properties:	
spatial resolutions	3-10km/1km
geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	regular latitude/longitude grid
Geographic Coverage:	Global/EU
Time span of output product:	SWI: Daily SWI10: 10-day period SWI TS (timeseries): 6 months
Timeliness of output delivery:	SWI: 12hrs SWI-10: within 3 days after synthesis period.
Uncertainty	**Threshold**: 0.1 m^3/m^3 Goal: 0.04 m^3/m^3
Stability	**Threshold**: 5% Goal: 3%

2.2 FURTHER REQUIREMENTS:

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land Component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

3 REVIEW OF INITIAL SWI1KM VALIDATIONS

In January 2020, a detailed Quality Assessment Report [CGLOPS1_QAR_SWI1km-V1] was produced to validate the Copernicus Land Monitoring Service's SWI1km product, which is based on Scatterometer-Synthetic Aperture Radar Soil Water Index (called SCATSAR-SWI) algorithm that ingests Surface Soil Moisture (SSM) from Metop ASCAT and Sentinel-1. For the period 2015-2019, the dataset was compared to in-situ data from suitable networks over Europe of the International Soil Moisture Network (ISMN). In the remainder of this section, the conclusions from the QAR are revisited.

Mostly confirming insights taken from recent studies (Bauer-Marschallinger et al., 2018; Termite et al., 2019; Azimi et al., 2020), the data fusion of Metop ASCAT SSM and Sentinel-1 SSM to the CLMS SWI1km using the SCATSAR-SWI algorithm is combining successfully the respective spatial- and temporal- signal quality of the two inputs and yields a value-added product. Data-driven fusion parameters are spatially consistent over the European area, with the exception of the Russian domain, where the Sentinel-1 data density is lower. Absorbing the spadework on the SWI, the approach on the data fusion of SSM retrievals from scatterometer- and SAR- sensors allows the generation of a daily SWI product at a kilometric scale, effectively combining at continental scale the high temporal frequency of the coarse SCAT input with the fine spatial details of the less-frequent SAR input. Common caveats of the coarse-scaled SCAT SSM data like biases along coasts or around cities are rectified through the data matching onto the SAR SSM patterns. In the present setup, the SCATSAR-SWI temporal dynamics are heavily relying on the SCAT component, simply due to the much larger observation number, particularly since Dec 2021 when Sentinel-1B observations ceased due to an anomaly[3].

The statistical time series comparisons with European in-situ soil moisture stations accessed via the ISMN showed an overall satisfying result, with performance comparable to the ASCAT-alone based product SWI V3 [CGLOPS1_VR2024_SWIV3], and much improved performance compared to the Sentinel-1-alone based product [CGLOPS1_VR2024_SSM1km].

[3] https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Mission_ends_for_Copernicus_Sentinel-1B_satellite

The level of agreement is almost constant for the different T-values / soil-depths. It varies more with the spatial domain, i.e. the agreement appears to be more dependent on the individual networks and their location (vegetation cover, soil type, climate, location's representativeness for its surrounding) or technological characteristics (sensor depth, sensor accuracy, stability). However, a clear relationship between soil depth and T-value and an assessment of potential time lags cannot be provided. Users are instead referred to recommendations following e. g. the analysis of Paulik et al. (2014).

As a prominent result from this evaluation, the time series analyses of the SWI1km reveal, for a remote sensing product, a relatively good skill to reproduce peaks and lows of the ground measurement's soil moisture at the near-surface layer. The data fusion approach seems to be most beneficial for capturing the temporal dynamics in the 24h-time-scale, resting on a dense SSM input signal fused from two Metop ASCAT and two Sentinel-1 sensors. This holds not true for the deeper layers, with less accurate reproduction of peaks and lows, and partially with time lags (delays) between satellite and in-situ time series. However, the general signal evolution, as well as the reduced variance of SM at such depths, is mostly well captured.

A seasonal bias appears at some stations, showing a general higher and lower level of soil moisture during spring and fall seasons, respectively. This issue seems to be inherited from the input ASCAT SSM and Sentinel-1 SSM datasets, which both suffer from a misinterpretation of vegetation density change and vitality dynamics in the radar backscatter modelling prior to SSM estimation (Hahn et al., 2017; Pfeil et al., 2018; Bauer-Marschallinger et al., 2018).

In cold climates, agreement with ground measurements is generally found to be rather low. Within the ongoing CLMS evolution activities, frozen surface states are currently investigated and a masking process based on meteorological data is developed to mask out irritated SSM retrievals caused by frozen surface conditions. This could improve the agreement of the next product version with ground measurements in colder climate regions.

4 SCIENTIFIC QUALITY MONITORING METHOD

In this study, the SWI1km product of the CLMS is evaluated on its signal quality and stability. The SWI1km retrieval is explained in detail in [CGLOPS1_ATBD_SWI1km-V1], and usage- and format-specifications are discussed in [CGLOPS1_PUM_ SWI1km-V1].

4.1 IN-SITU REFERENCE PRODUCTS

For in-situ validation purposes, the project consortium can access the comprehensive database of the ISMN (International Soil Moisture Network; Dorigo et al., 2011, Dorigo et al, 2021). It is an international cooperation to establish and maintain a global in-situ Soil Moisture (SM) database, and is recommended by CEOS' Soil Moisture Product Validation Good Practices Protocol (Montzka et al., 2020).

This database is an essential means of the geoscientific community for validating and improving global satellite observations and land surface models. As of March 2025, filtered for the period 2015 to 2024, the ISMN contains data of 85 networks and more than 4000 stations located all around the globe, with highest density over Europe and North America and most datasets developed during the last two decades. The database is rapidly expanding, which means that both the number of stations and the period covered by the existing stations are still growing. Hence, it will become an increasingly important resource for validating and improving satellite-derived soil moisture products and studying climate related trends. As the ISMN is supported by the scientific community itself, potential networks are invited to enrich the collection by sharing its in-situ soil moisture data.

The ISMN employs an automated quality control system for its SM measurements, which flags values exceeding a certain threshold and checking the validity of SM variations in relation to changes in soil temperature and precipitation (Dorigo et al., 2013). A second flagging procedure analyses the shape of the SM time series to detect outliers, positive and negative breaks, saturation of the signal, and unresponsive sensors. In this analysis, only values with the flag "G" (good) are kept. Dorigo et al. (2021) reported on the recent developments within the ISMN and provide a detailed documentation on the network update activities and data quality curation. This includes an ongoing activity on quality curation of historic time series that aims for most consistent and reliable in situ data. Consequently, the available time series data at ISMN may be altered over time in terms of flagging and value density. To ensure a long-term funding, the ISMN operations were transferred by the end of 2022 to the International Center for Water Resources and Global Change (ICWRGC) and the German Federal

Institute of Hydrology (BfG), and both new hosting intuitions are dedicated to continue ongoing activities on network integration and quality check.

4.1.1 Used ISMN Networks

For the validation of the SWI1km dataset, the ISMN was browsed for networks that fulfil the following requirements:

- Location in the SWI1km coverage area, i.e. the Europe production window
  - bottom left: -12°E, 35°N, top right: 45°E, 72°N
- Valid observations during the validation period Jan 2015 – Dec 2024
  - First, a suitable station needs at least one year of in-situ data available to be considered
  - After temporal matching with SWI1km and masking of cold temperatures, at least 30 observations per station must be available.
  - As further criteria on the ISMN input data density, results in section 5 are only plotted with data from locations, where the correlation has a high significance with a p-value lower than 0.05.
- SM measurements at the near-surface, subsurface and deep layers, according to depths listed in Table 2.

The ISMN networks are constantly updated, either via an Application Programming Interface (API) automatically refreshing the data in near-real-time, or via manual updates. Some networks have update delays and were not updated for entire 2024 at the time of the validation. The ISMN data was accessed the 15th of April 2025. Comparing to last year reports, the addition of a German network with Sentek Drill and drop sensors can be highlighted (DWD). This network provides in-situ soil moisture (between 5-85 cm) with Sentek Drill and drop in 20 locations. While the two networks (COSMOS and COSMOS-UK) did not get updated for 2024, their use of cosmic ray neutron sensors provides higher accuracy data for comparison with SWI1km product.

The validation of the SWI1km is twofold: 1) For the long-term validation, every sensor with data between 2015 and (inclusive) 2024 are used; 2) when comparing the product quality for 2024, the networks and sensors are filtered to only use the ones with measurements available for the period of 2024, leading to a smaller set of analysed networks.

A detailed overview of the locations of all used ISMN stations fulfilling the requirements delineated above is shown in Table 1. They are distributed over the continent as shown in Figure 1

and thus represent very well the area covered by SWI1km. For some networks, a larger number of stations than indicated in the "suitable stations" column is available via ISMN. However, some of these stations only measured data for a few weeks, which hampers the robust calculation of validation metrics. In order to ensure a reliable derivation of evaluation results, only stations with at least 30 collocated SWI and in-situ observations are kept. This follows the suggestion drawn from the Central Limit Theorem of probability theory for parametric tests like Pearson correlation.

For more information on the ISMN network details and contacts, please visit: https://ismn.earth/en/networks/.

Table 1: ISMN networks used in the present SWI1km validation campaign with characteristics as given in the ISMN documentation. At some stations, multiple sensors are installed at the same depth. Each of these sensors is included in the number of suitable stations.

Network Name	Data Availability	Climate (Koppen-Geiger) *	Altitude [min-max] m	Suitable Stations Near-Surface	Suitable Stations Sub-surface	Suitable Stations Deep Soil
BIEBRZA_S-1 Poland	2015-04-23 to 2018-12-01	Dfb	110-112	29	24	5
COSMOS Austria/Switzerland	2010-12-13 to 2017-04-05	Cfb, Dfb	270-750	2	4	0
COSMOS-UK Great-Britain	2013-03-10 to 2023-01-01	Cfb	3-365	137	263	39
DWD Germany	2024-01-01 to 2024-12-31	Cfb, Dfb	3-877	19	38	78
FMI northern Finland	2007-01-25 to 2024-12-31	Dfc	150-441	37	25	13
FR_Aqui France	2012-01-01 to 2023-12-31	Cfb	25-54	8	11	14
GROW Europe	2017-02-08 to 2019-10-08	Cfb, Dfb	European network	142	0	0
HOAL Austria	2013-07-11 to 2021-12-31	Dfb	270-321	71	68	31
HOBE Denmark	2009-09-08 to 2019-03-13	Dfb	26-104	141	139	141
HYDROL-NET-PERUGIA Italy	2010-01-01 to 2016-12-31	Csa	285	2	4	2

IMA_CAN1 **Italy**	2011-08-23 to 2015-12-03	Cfa	272-278	11	12	0
IPE **northern Spain**	2008-04-03 to 2020-03-25	Cfb	750	1	1	0
LABFLUX **Italy**	2020-04-23 to 2023-06-19	Dfb	640	1	2	1
MOL-RAO **eastern Germany**	2003-01-01 to 2020-06-30	Dfb	49-73	2	5	4
NVE **Norway**	2006-07-12 to 2024-12-31	Dfb, Dfc, ET	68-950	5	14	12
REMEDHUS **central Spain**	2005-03-15 to 2024-12-31	Bsk	700-900	20	0	0
RSMN **Romania**	2014-04-09 to 2024-12-31	Cfa, Dfa, Dfb, Dfc	19-318	20	0	0
SMOSMANIA **southern France**	2007-01-01 to 2024-01-01	Cfa, Cfb, Cfc, Csa, Dfa	30-1240	56	80	24
STEMS **northern Italy**	2015-12-04 to 2024-12-31	Cfa	275	4	11	7
TERENO **western Germany**	2009-12-31 to 2024-12-31	Cfb	70-612	14	14	14
TWENTE **Netherlands**	2008-19-11 to 2020-12-31	Cfb, Dfb	0-49	74	71	32
UMBRIA **Italy**	2002-10-09 to 2017-12-31	Csa, Cfa	215-550	9	9	9
UMSUOL **Italy**	2009-06-12 to 2017-05-15	Cfa	11	1	2	2
WATERLINE **Greece**	2021-10-06 to 2024-09-09	Csb	34-711	2	0	0
WEGENERNET **Austria**	2007-01-01 to 2024-12-31	Dfb	265-398	0	25	3
WSMN **Great Britain**	2011-09-02 to 2016-02-29	Cfb	28-477	4	2	0
XMS-CAT **eastern Spain**	2016-08-01 to 2024-12-31	Dfb, Dfc, Csb	38-1236	16	17	19

* Koppen-Geiger climate information:
1st letter : B (dry), C (temperate), D (Continental), E (Polar)
2nd letter : w (Dry winter), f (No dry season), s (Dry summer),
3rd letter: a (Hot summer), b (Warm summer), c (Cold summer), k (Cold for dry climate only), T (Tundra)

4.1.2 Soil Depth Levels

The depth levels for the present study are defined in Table 2. The associated T-values of the SWI products are selected in accordance with the qualitative assessment summarised in the Appendix "Recommendations for T-values and Soil Depth" of [CGLOPS1_QAR_SWI1km-V1], and are in agreement with the guidelines discussed by Paulik et al. (2014). In the annual quality monitoring analyses for the ASCAT SWI V3 (as e.g. reported to [CGLOPS1_VR2024_SWIV3]), a strong correlation between evaluation results from SWI data with different T-values was found and data quality and integrity can be evaluated with a subset of T-values. In order to keep this document concise and reduce redundancy, only the SWI datasets with T-values 2, 10 and 100 are analysed in this study. Figure 1 shows the locations of all used ISMN stations fulfilling the requirements delineated in 4.1.1 for the validation of the SWI data (at T=2, 10 and 100).

Table 2: Depth levels for validation experiments.

Level	Depth [m]	T-value
Near-Surface	0.00 - 0.10	2
Sub-Surface	0.10 - 0.30	10
Deep Soil	0.30 - 0.80	100

© C-GLOPS Lot1 consortium

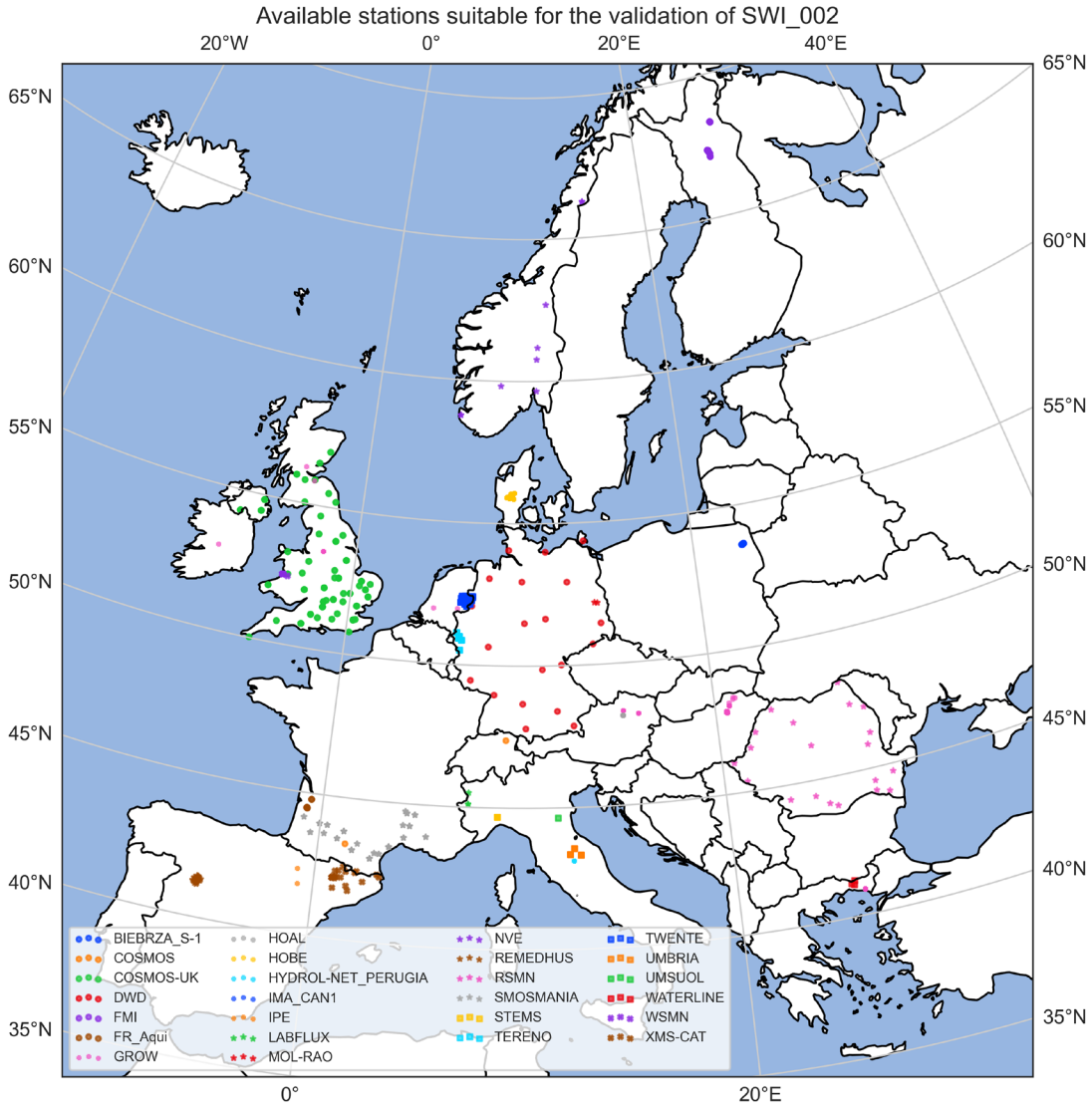


Figure 1: Locations of input ISMN stations (colored dots) for the 2024's SWI1km analysis available at the near-surface.

© C-GLOPS Lot1 consortium

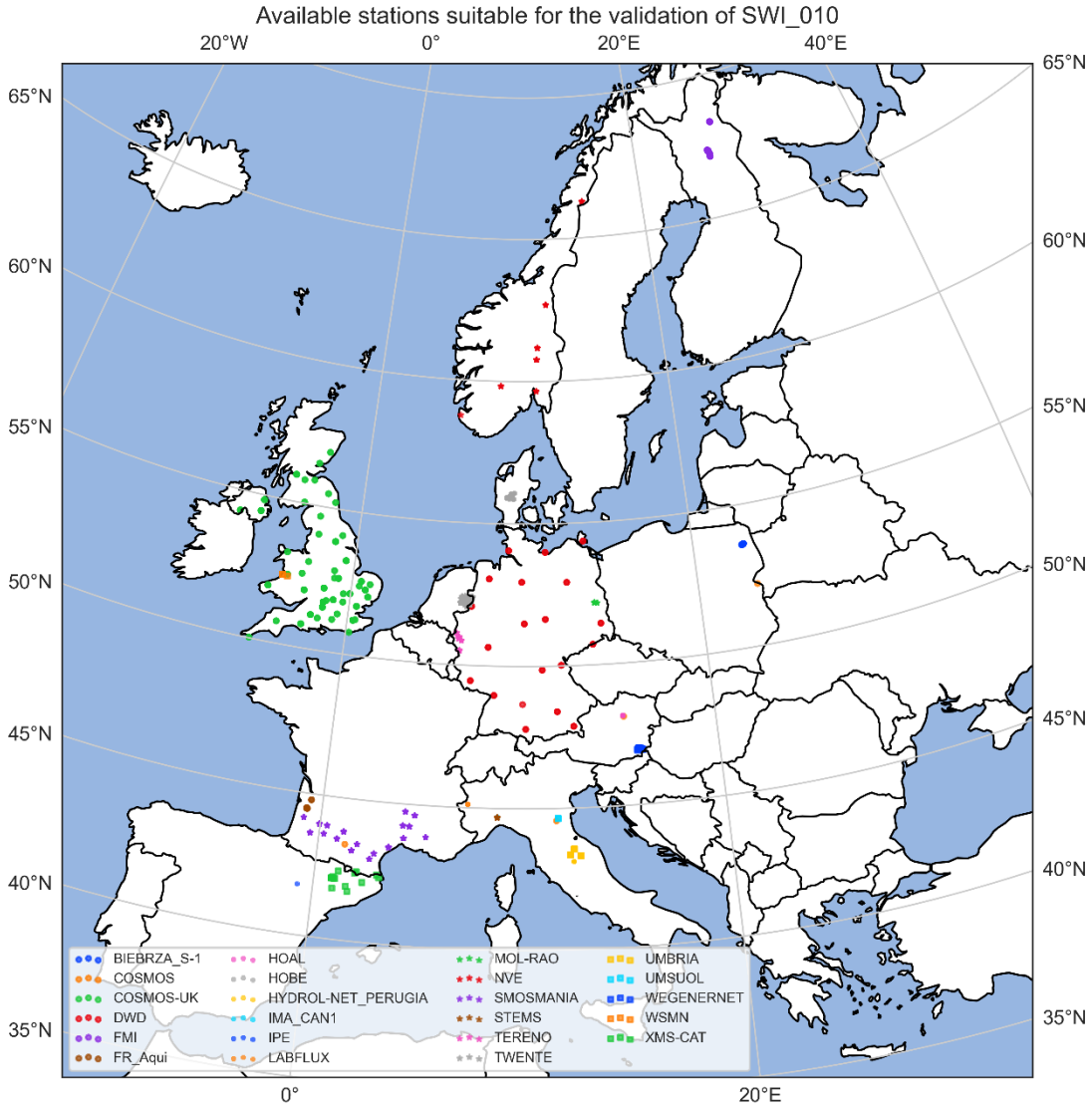


Figure 2: Locations of input ISMN stations (colored dots) for the 2024's SWI1km analysis available at the sub-surface.

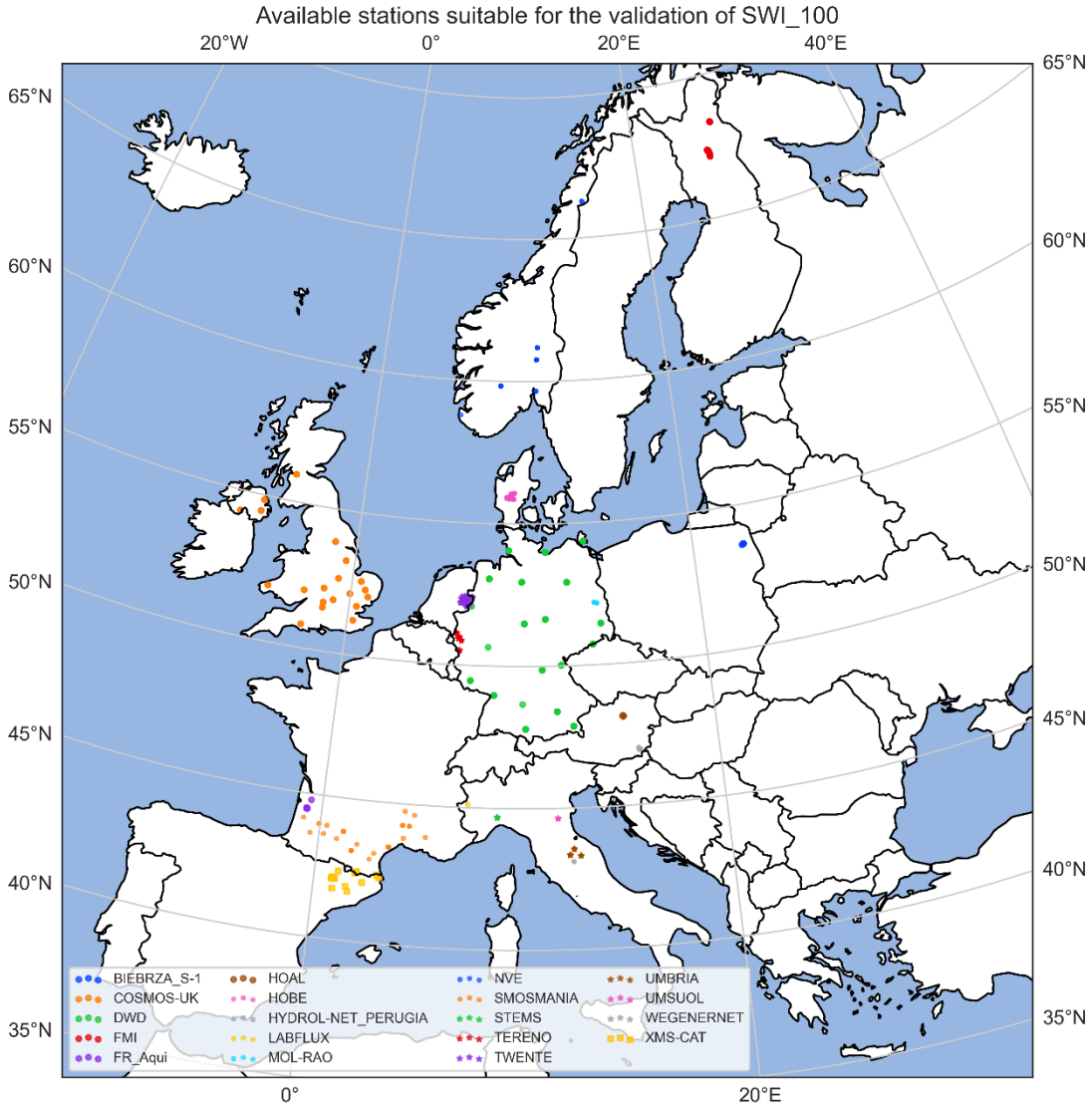


Figure 3: Locations of input ISMN stations (colored dots) for the 2024's SWI1km analysis available at the deep soil.

© C-GLOPS Lot1 consortium

4.1.3 Spatial Representativeness

For the validation of high-resolution soil moisture products as the SWI1km, actual ground measurements from in-situ stations are of major importance. Such measurements are able to represent soil moisture variations at point- or field-scale, depending on the location's area homogeneity and the sensor design. When comparing remotely sensed and in situ data, one must consider spatial representativeness. This means how well an in-situ observation, which is essentially a point measurement, represents the data gathered by the much larger satellite footprint. This is in general different for each station and depends on topography, soil types and microclimates. De Lannoy et al. (2007) discussed relationship between point measurements and the average behaviour of field-scale soil moisture and concluded that the choice of individual sites has considerable impact on validation campaigns, suggesting field-level analysis of the local spatial SM signal structure and its drivers, followed by upscaling procedures.

For this validation, all datasets are kept with their native product samplings as an in-depth analysis of in-situ sites is out of scope in this endeavour. Consequently, results from this validation need always be understood in reflection of the large difference of scales. In the case of the present validation study of 1km data, this so-called representativeness error is assumed not negligible and thus validation results need to be reflected accordingly. Crow et al. (2012) stated that during validation activities based on comparisons between point-like ground observations and extensive satellite retrievals, this sampling error can be misattributed to retrieval uncertainty and spuriously degrade the perceived accuracy of satellite soil moisture products. They found that standard deviations of SM in sample sets spanning 0.8km² and 1.6km² are of the same or larger magnitude as the CEOS' accuracy criterion for SM of 0.04 m³/m³. Gruber et al. (2013) carried out an in-depth analysis of coarse-scale representativeness of ISMN networks and they found absolute error estimates ranging from 0.02 to 0.06 m³/m³. However, one can expect general lower errors when analysing SM anomalies and relative dynamics.

Alternatively, a validation against modelled SM data of the same scale as the satellite product could be done. However, for a continentally or globally comprehensive validation of remotely sensed SM at kilometric scale, there is a lack of comparable extensive datasets with similar spatial resolution and extent. Fortunately, regional model output at kilometric scale describing SM, even with limited extent, can provide an accurate assessment of the temporal and spatial SM signal quality. The reader may here refer to the study of Bauer-Marschallinger et al. (2018) where the authors examined a

SWI1km dataset against an extensive 1 km soil moisture model dataset covering the Umbria region in Italy.

4.2 VALIDATION PROCEDURE

The aim of this validation report is to assess the performance of the SWI1km product in the period from 2015-01-01 to 2024-12-31 with respect to ground measurements at the stations collected by the ISMN. This is done for three soil depth levels to evaluate the behaviour in different soil layers: 1) near-surface, 2) sub-surface, and 3) deep soil. For each layer, SWI time series with corresponding T-value are extracted and statistically compared to time series from in-situ stations at the same location. All available in-situ time series measured at the corresponding depth level are considered in the statistical analysis.

4.2.1 Dataset Preparations

4.2.1.1 Temporal Resampling

The SWI1km datasets are available with a daily timestamp. The reference in-situ datasets offer generally a much higher sampling and are thus down-sampled for this validation to daily mean values. Thereby, also potential diurnal jamming of in-situ soil moisture sensors through e.g. temperature effects are removed. The daily SWI1km images are converted to time series through co-registered stacking along the time dimension.

4.2.1.2 Masking for Invalidating Conditions

SWI1km and in-situ time series are temporally masked for invalid observations, where soil moisture measurements are prone to be irritated. Typical conditions are frozen soil and snow. In this analysis, we use the quality flagging of the ISMN in-situ data, the flags for snow coverage, frozen soil, saturated signal plateaus, and excessive precipitation (see Section 4.1).

The ISMN team announced on January 16th, 2025 that the algorithm for data quality flagging was updated and improved. These updates mainly affected three of the quality flags (D03, D05, and D10) on the following NRT networks (FMI, RSMN, TERENO) and the regular networks (COSMOS, FR_Aqui, HOAL, LABFLUX, REMEDHUS, STEMS, and WATERLINE) used in our analysis. It should be noted that the changes in flagging of ISMN data might have affected the values in the data

for years previous to 2024. However, only marginal changes can be seen in the statistical values (rho of 0.09 and RMSD of 0.02). It should be noted that a changing sample size due to ISMN flag updates (e.g. from updated frozen flagging, saturated plateaus, snow or excessive precipitation) – and related shifts in mean and standard deviation – can cause different appearance of the scaled SWI1km data in the different annual QAR reports, accordingly (or even alter the decision if the location falls below the threshold of 30 matching observations).

4.2.1.3 Temporal Matching

For the time series comparison, the individual datasets are checked at each validation location for no-data-values. Then both time series are time-stamp-matched, reducing them to timestamps where both time series have valid data. Two timestamps are matched if they co-located within a 12 hours temporal interval. This allows for the calculation of correlation coefficients and Root Mean Square Difference (RMSD) values as measures of agreement.

4.2.1.4 Data Scaling

Since the SWI1km data is provided in percentage of relative saturation (i.e. %), while the reference products are provided in m^3/m^3 (ranging commonly from 0.0 – 0.5 m^3/m^3), data scaling is necessary prior to the evaluation of difference metrics such as RMSD. Following leading experts and current literature, scaling into the in-situ data space is done (Brocca et al., 2011, Montzka et al, 2020).

In order to report here the targeted metrics in m^3/m^3, “mean-standard-deviation scaling” (also known as z-normalisation) is used to scale the SWI data to the reference products (Lima and Souza, 2023). With this method, the SWI data is scaled to have the same mean value and standard deviation as the reference dataset. This approach was chosen because it is widely adopted by the community and because the extent of the scaled values is less affected by the presence of outliers, compared to for instance a “minimum-maximum scaling”. Moreover, through the mean-standard-deviation scaling, no statistical bias remains between CGLS and reference time series, and the obtained RMSD values are equivalent to the unbiased Room Mean Square Error (ubRMSE), a metric regularly used within good practices established by the CEOS LPV (Montzka et al, 2020).

4.2.2 Assessment of the Temporal Signal Quality

For the assessment of the temporal signal quality, the SWI1km data is compared with data from the selected ISMN networks. For the comparison, basic statistical measures are calculated: the Pearson correlation coefficient (rho) and the Root Mean Square Difference (RMSD).

The obtained validation metrics are presented and discussed in Section 5. First, an overview of the overall SWI1km dataset quality is given per network, along with an assessment of quality differences between the SWI1km dataset in the year 2024 and in the reference period 2015-2023. This is complemented by an in-depth analysis over selected individual ISMN stations, discussing the temporal dynamics of SWI1km and in-situ SM time series and their agreement. All analyses are carried out for three soil layers (near surface, sub-surface, deep soil).

© C-GLOPS Lot1 consortium

5 RESULTS

This section first presents the mean values of the performance indicators Pearson rho and RMSD per network, then shows summary box plots and network overviews per soil depth layer, and features in the end time series plots to illustrate selected examples of good and poor agreement between SWI1km and in-situ.

5.1 OVERALL ANALYSIS SUMMARY

The results of the statistical comparison of SWI1km against ISMN in-situ stations are overall satisfying, with consistent performance over different soil depth layers (Table 3). The weighted means (with the number of stations used as weight) show similar correlation for all three depths.

The results for the deep soil layer are calculated over fewer in-situ stations and show lower Pearson correlation compared to shallower depths. The results are comparable to the results obtained for the period 2015-2023 in the last year experiment, and also with the initial evaluation over the period 2015-2019 [CGLOPS1_QAR_SWI1km-V1]. The RMSD values for all layers maintained similar values with the addition of 2024 data.

Overall, the average **rho of 0.51-0.57** and overall average **RMSD of 0.05-0.07** m^3/m^3 are considered satisfying, in respect to the expected representativeness errors between the in-situ point measurements and the SWI's 1km scale. For certain networks, the dataset quality is considerably better than the average. For example, the maximum average rho of 0.78 (near surface) over WATERLINE, 0.73 (sub-surface) over IMA-CAN1 and TWENTE, and 0.82 over UMSUOL (deep layer). Along with minimum RMSD values of 0.03 m^3/m^3 for IMA-CAN1 at near surface, IMA-CAN1 and IPE at subsurface, and 0.02 m^3/m^3 for DWD for the deep soil layer. It should be further noted that the performance also varies among the stations of each network due to differences between the specific sites, e.g., related to land cover and soil composition. The results of the statistical comparison are presented in more detail in the following sections.

Please note that the values in Table 3 are mean values, whereas the line, box and whiskers in the box plots (next sections) are derived from median values and percentiles. The obtained scores are higher than for the Sentinel1-alone SSM1km product (with the exception of the network: FMI; see [CGLOPS1_VR2024_SSM1km]), suggesting that the SWI1km profits much from the ASCAT SSM input signal quality and temporal density. Moreover, the obtained scores are higher than the results obtained from the well-established (global) ASCAT-alone SWI V3 product at 12.5 sampling (see

[CGLOPS1_VR2024_SWIV3]). However, the comparison should be treated cautiously as the SWI T-values are different (for SWIV3: T=1/20/100 and for SWI1km: T=2/10/100), and the temporal analysis covers 2007 to 2024 for SWIV3 resulting in a different set of ISMN networks and only partial overlapping. This nonetheless suggests that the data fusion and the combined SWI generation from Sentinel-1 SSM and ASCAT SSM are working correctly and even improve ASCAT's temporal signal quality (in addition to improving the spatial resolution). Hence, the SCATSAR-SWI data fusion approach can be considered as an improvement that leads to a higher capability to operationally monitor SM. This is also supported by the analysis of Bauer-Marschallinger et al. (2018), where the authors compared SCATSAR-SWI data over Italy against SM data from in-situ and rainfall data from weather radars, as well as against 1km model data over Umbria, and found similar improvements.

Table 3: Overall summary of statistics from the time series analysis of SWI1km versus in-situ measurements: Mean values of Pearson rho and RMSD, per network, and total per soil layer. The number of included in-situ stations is given by n.

Network	Near Surface			Sub-Surface			Deep Soil		
SWI1km's T-Value	**T=2**			**T=10**			**T=100**		
	n	Ø rho	Ø RMSD [m³/m³]	*n*	Ø rho	Ø RMSD [m³/m³]	*n*	Ø rho	Ø RMSD [m³/m³]
BIEBRZA_S-1	*29*	0.31	0.07	*24*	0.33	0.07	*5*	0.10	0.06
COSMOS	*2*	0.46	0.05	*4*	0.46	0.04	*0*		
COSMOS-UK	*137*	0.65	0.07	*263*	0.70	0.07	*39*	0.59	0.05
DWD	*19*	0.57	0.07	*38*	0.56	0.04	*78*	0.58	0.02
FMI	*37*	0.00	0.06	*25*	-0.12	0.06	*13*	-0.12	0.05
FR_Aqui	*8*	0.71	0.04	*11*	0.70	0.04	*14*	0.61	0.05
GROW	*142*	0.54	0.08	*0*			*0*		
HOAL	*71*	0.45	0.08	*68*	0.43	0.08	*31*	0.45	0.06
HOBE	*141*	0.52	0.05	*139*	0.57	0.05	*142*	0.54	0.05
HYDR-NET_PERUGIA	*2*	0.61	0.06	*4*	0.63	0.04	*2*	0.64	0.04
IMA_CAN1	*11*	0.65	0.03	*12*	0.73	0.03	*0*		
IPE	*1*	0.29	0.04	*1*	0.38	0.03	*0*		
LABFLUX	*1*	0.61	0.04	*2*	0.56	0.05	*1*	0.09	0.06
MOL-RAO	*2*	0.52	0.05	*5*	0.44	0.04	*4*	0.65	0.03
NVE	*5*	0.60	0.04	*14*	0.28	0.05	*12*	0.30	0.03

REMEDHUS	*20*	0.66	0.05	*0*			*0*		
RSMN	*20*	0.51	0.04	*0*			*0*		
SMOSMANIA	*56*	0.54	0.05	*80*	0.51	0.06	*24*	0.35	0.06
STEMS	*4*	0.68	0.07	*11*	0.66	0.04	*7*	0.62	0.06
TERENO	*14*	0.60	0.06	*14*	0.56	0.06	*14*	0.61	0.05
TWENTE	*74*	0.70	0.07	*71*	0.73	0.06	*32*	0.68	0.05
UMBRIA	*9*	0.58	0.06	*9*	0.62	0.05	*9*	0.37	0.06
UMSUOL	*1*	0.63	0.06	*2*	0.68	0.06	*2*	0.82	0.06
WATERLINE	*2*	0.78	0.08	*0*			*0*		
WEGENERNET	*0*			*25*	0.38	0.08	*3*	0.53	0.07
WSMN	*4*	0.63	0.05	*2*	0.65	0.04	*0*		
XMS-CAT	*16*	0.53	0.09	*17*	0.45	0.08	*19*	0.39	0.06
Weighted Mean (by number of stations); *total number of stations*	*828*	**0.54**	**0.07**	*841*	**0.57**	**0.06**	*450*	**0.51**	**0.05**

5.1.1 Summary over Networks: Pearson Correlation Analyses

In the following, box plots for the investigated SWI-T-values (2, 10, 100) show the distribution of the Pearson rho values per network for the evaluation period 2015-2024 (Figure 4, Figure 7, Figure 10). Only the stations with a significant result and a Pearson p-value below 0.05 are used for the box plots. Please note, in the figures, the number of contributing stations is described by “n”, and that in case of a single station box plots are replaced by horizontal lines.

In general, for the individual layer’s analysis, the observed scores remain similar, compared to last year analysis. For the near-surface layer (Figure 4), most station-networks show relatively good scores with the majority of median rho values ranging from 0.40 to 0.75. All median rho values except for the BIEBRZA-S-1, FMI, and IPE networks are above 0.40. The lower values for these three networks can be explained by challenging land covers (wetlands for BIEBRZA_S-1, dense forest for IPE and FMI) where SWI estimates from ASCAT and Sentinel-1 are less reliable. Moreover, the FMI network is located at a cold climate (northern Finland), leading to large data gaps due to frozen periods resulting in low to negative correlation over the Finnish network. Similar trend can be observed for the sub-surface and deep soil layers (Figure 7, Figure 10). However, deep soil layer shows a higher percentage of networks around and below 0.4 rho values, compared to subsurface soil layer.

The analysis of the 2015-2024 period is complemented by a comparison of the Pearson rho values obtained when using only data from the year 2024, versus the obtained rho values for the reference period 2015-2023. Only networks that measured soil moisture consistently in 2024 (with minimum 30 observations and a p-value below 0.05 for 2024) have been included in the analysis, narrowing the set of analysed stations. The box plots for near-surface (Figure 5) shows improving correlation in 2024 with respect to the reference period for the networks FMI, REMEDHUS, WATERLINE, and XMS-CAT. The TERENO network correlation remains high with a median correlation above 0.55 for 2024. Over the Romanian network RSMN, the median correlation value drops for 2024 by a margin of 0.2. This might be explained because only a fraction of the stations is still operated and some of them with incomplete time coverage. Furthermore, the correlation distribution over RSMN is characterised by large spread of values indicating variability between analysed stations. Over the Italian network of STEMS, a median correlation drop of 0.15 can be seen for the year 2024, however the correlation median value stays high at 0.65. Only 2 out of the 4 stations have been updated, and both are missing data for multiple months in year 2024. The subsurface layer box plot (Figure 8) shows similar or improved Pearson coefficient for the networks FMI, WEGENERNET, and XMAS-CAT. The WEGENERNET network shows overall correlation increase (from rho = 0.37 to 0.4). However, XMAS-CAT shows the greatest increase (from rho=0.37 to rho=0.67), where many more stations with data in 2024 compared to 2023 have been added to the network and the quality of data in 2024 has increased. However, we see reduced Pearson coefficients for STEM, TERENO, and NVE. These lower values are potentially explained by longer period of masked data due to the continental climate and cold winter. As for the network NVE, only one station collected data for the year 2024, and it still had many missing months of data. Finally, the results over FMI might be explained by the large data gaps caused by frozen masking, reducing the temporal availability of soil moisture measurements, as well as by the challenging land cover type. The change in correlation is considered more likely driven by temperature or vegetation geometry changes than by lower soil moisture sensitivity. SWI at T=100 maintains a satisfactory Pearson correlation against the available in-situ networks for 2024 when compared to the reference period (Figure 11). TERENO shows a significant drop that can be explained by some stations not having data for 2024 and even out of the ones with values in 2024, many have large missing sections. Western Germany (where the TERENO network is located) had the rainiest year in recorded history in 2024. This is in contrast to XMS-CAT where more and higher quality data was available in 2024 compared to previous years. Compared to the subsurface layer, the correlation is slightly lower and with a higher standard deviation due to the nature of the SWI approach.

Finally, the Pearson correlation values are shown as maps over selected networks (Figure 6, Figure 9, Figure 12). One can see that the agreement with the SWI1km dataset varies also among stations of the same network. This variation is in some cases similar to what is observed for the SSM1km surface soil moisture product (e. g. for HOAL, BIEBRZA-S-1, TERENO, REMEDHUS; see [CGLOPS1_VR2024_SSM1km]). For the other networks, there are some spatial differences within the correlation values within each network. This can most likely be explained by the inclusion of the temporally dense ASCAT signal into the SWI1km product as well as stations covering varying land covers.

5.1.1.1 Near-surface Soil Layer (T-Value = 2)

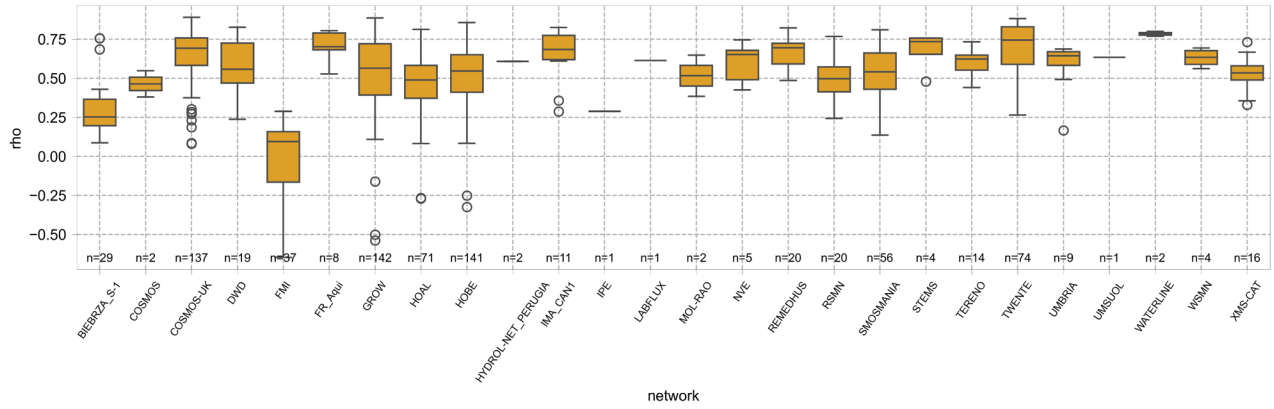


Figure 4: Boxplot summarizing Pearson rho values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) for the period between 2015 and 2024, per network. Median values are represented by the horizontal line. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points.

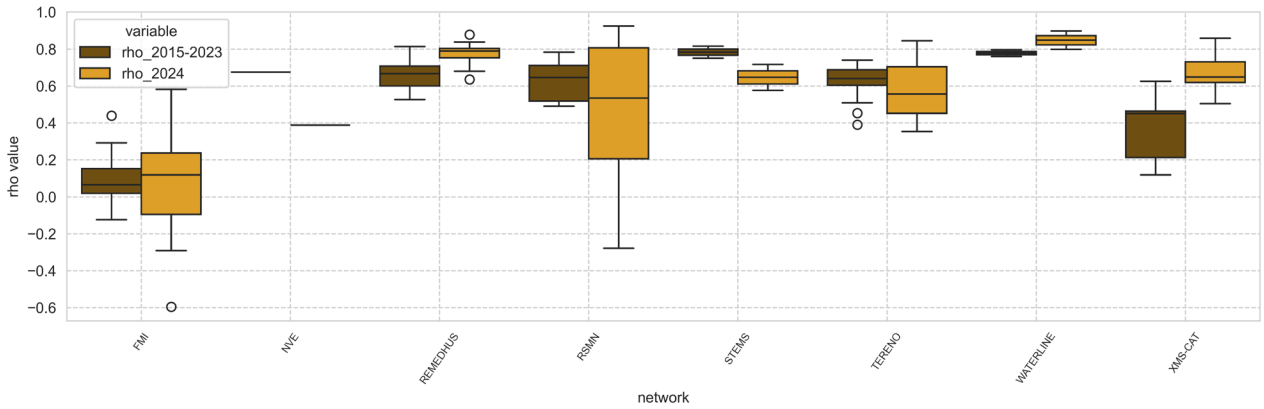


Figure 5: Box plot summarizing Pearson rho values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) per network, based on data from 2015-2023 (brown) and 2024 (orange). Only networks with data covering 2024 are included. The boxes indicate 25th and 75th percentiles, the whiskers are set to 5th and 95th percentile and outliers are indicated by individual points.

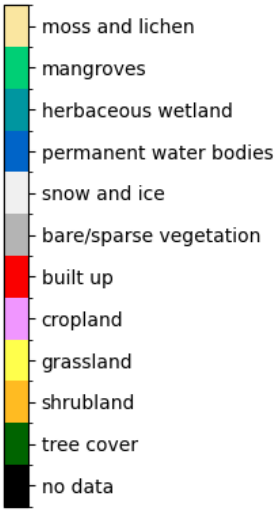


Table 4: Background land cover types for Figure 6 to Figure 12, derived from the ESA WorldCover map 2020.

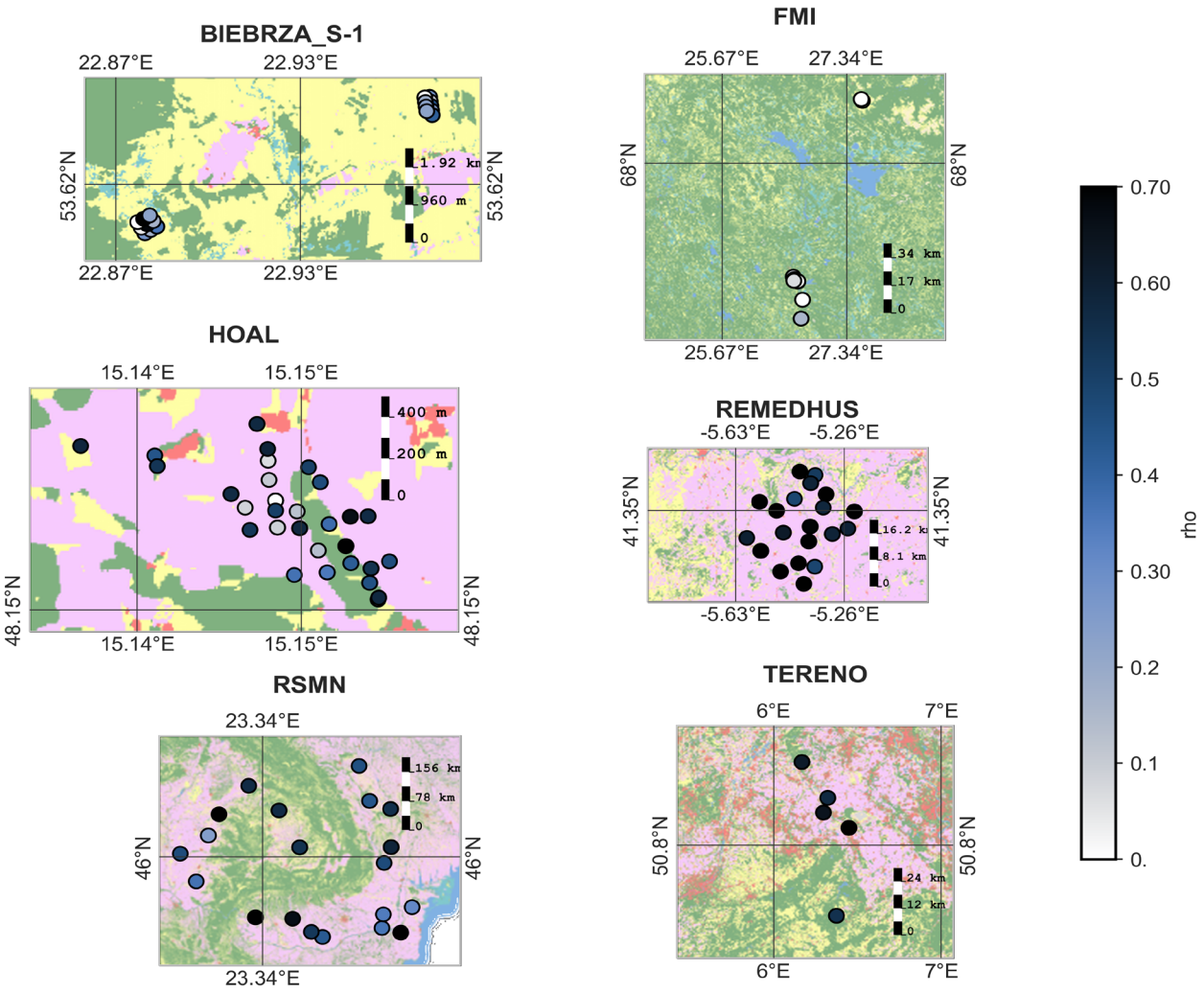


Figure 6: Pearson correlation (rho) for SWI1km with T-value=2 over selected ISMN networks (0-10 cm depth). See the land cover classes in Table 4.

5.1.1.2 Sub-surface Soil Layer (T-Value = 10)

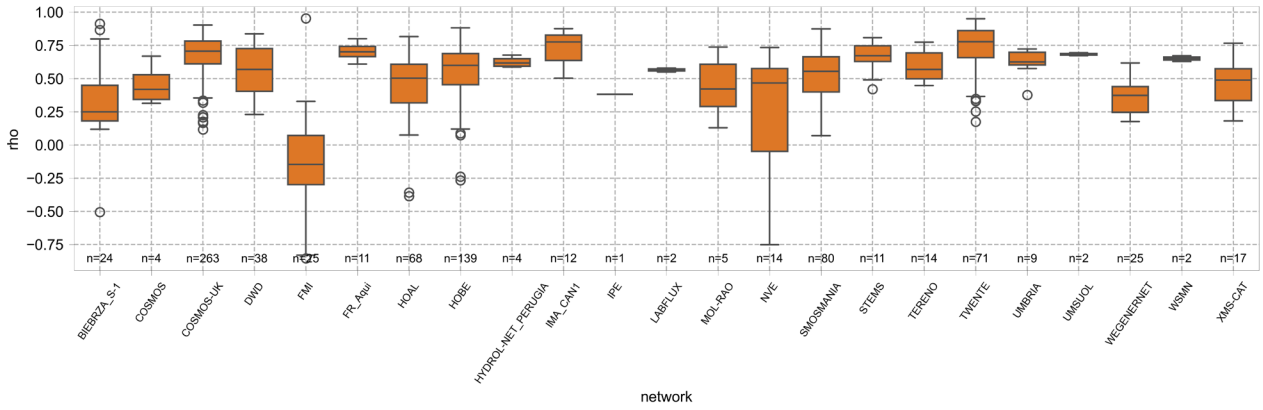


Figure 7: Boxplot summarizing Pearson rho values calculated for SWI1km with T-value=10 and all relevant ISMN stations (10-30 cm depth) for the period between 2015 and 2024, per network.

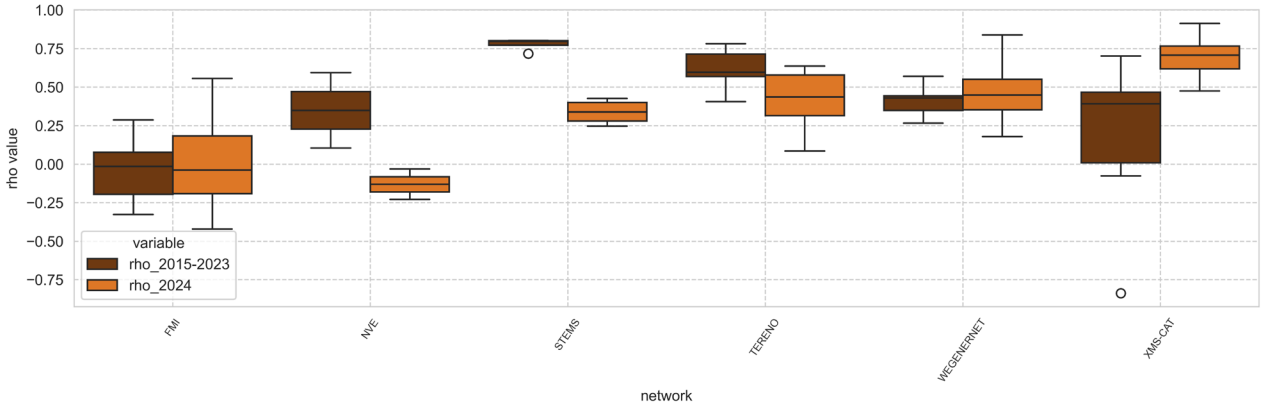


Figure 8: Box plot summarizing Pearson rho values calculated for SWI1km with T-value=10, and all suitable ISMN stations (10-30 cm depth) per network, based on data from 2015-2023 (brown) and 2024 (orange). Only networks with data covering 2024 are included.

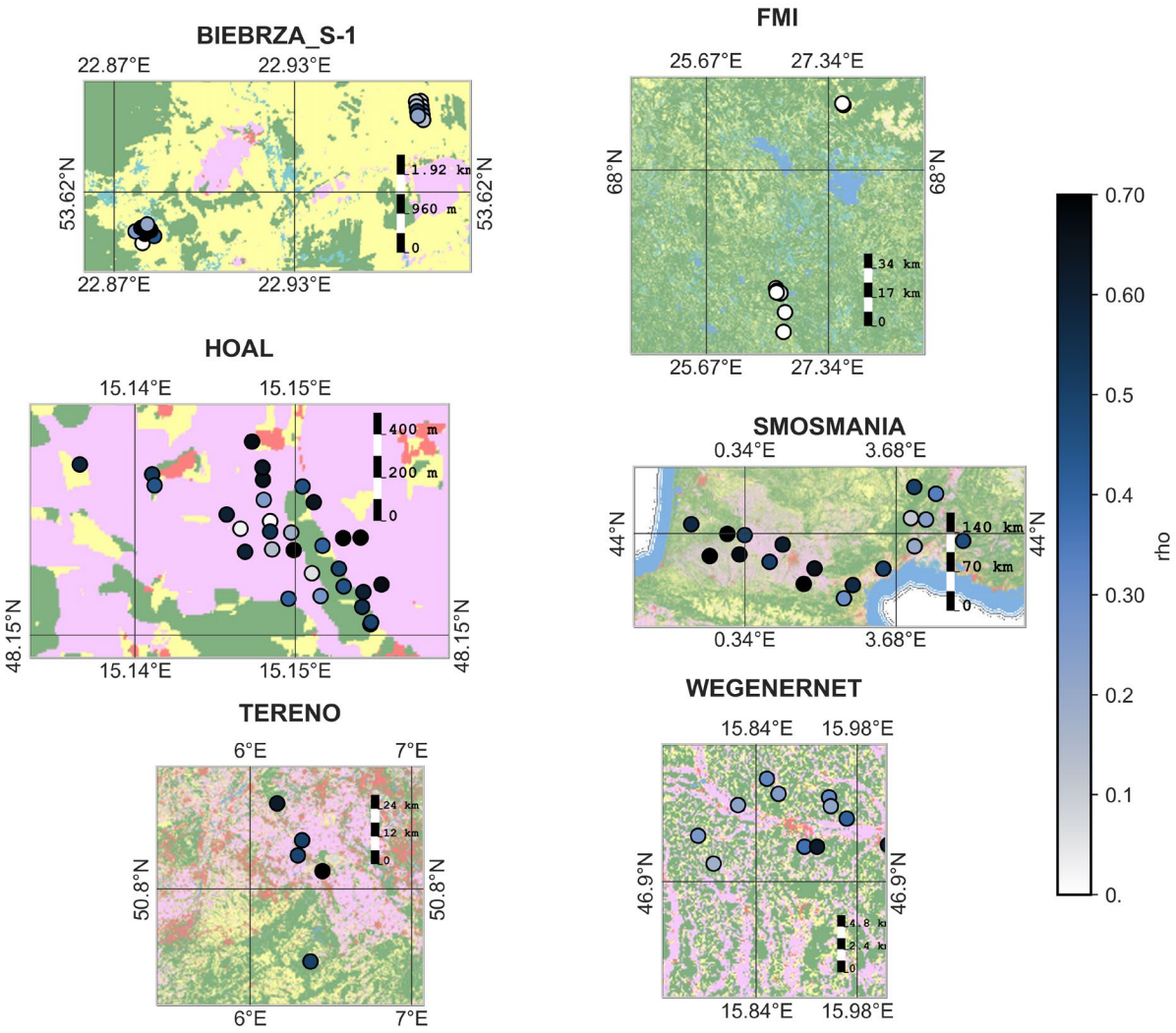


Figure 9: Pearson correlation (rho) for SWI1km with T-value=10 over selected ISMN networks (10-30 cm depth). See the land cover classes in Table 4.

© C-GLOPS Lot1 consortium

5.1.1.3 Deep Soil Layer (T-Value = 100)

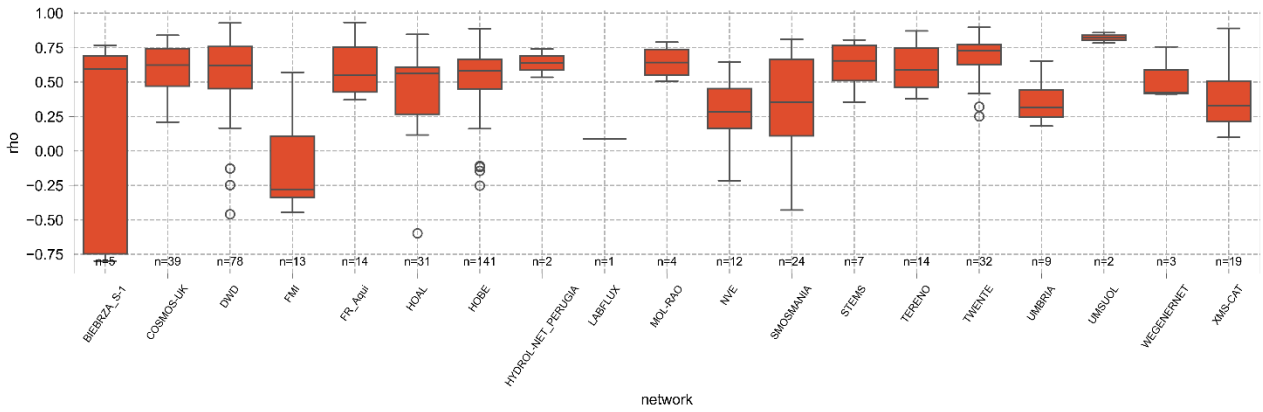


Figure 10: Boxplot summarizing Pearson rho values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) for the period between 2015 and 2024, per network.

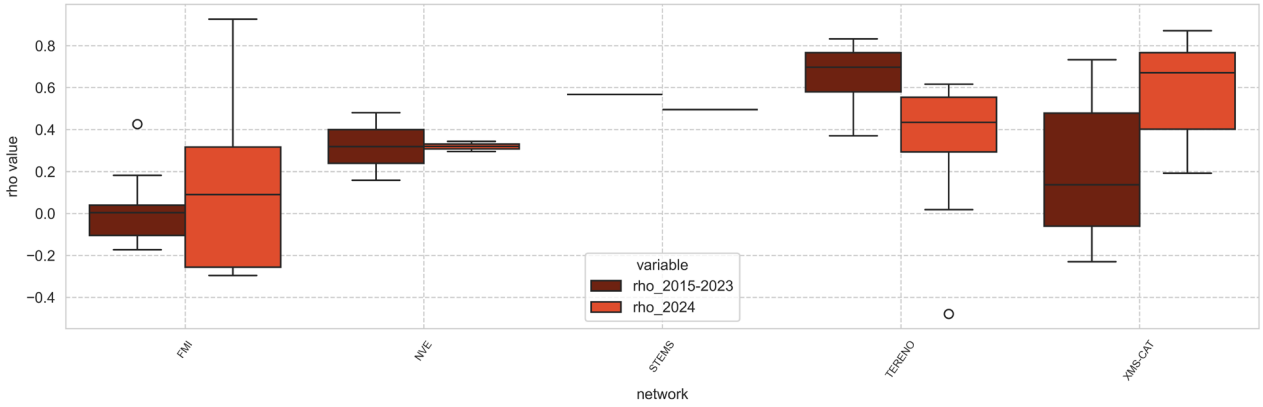


Figure 11: Box plot summarizing Pearson rho values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) per network, based on data from 2015-2023 (brown) and 2024 (red). Only networks with data covering 2024 are included.

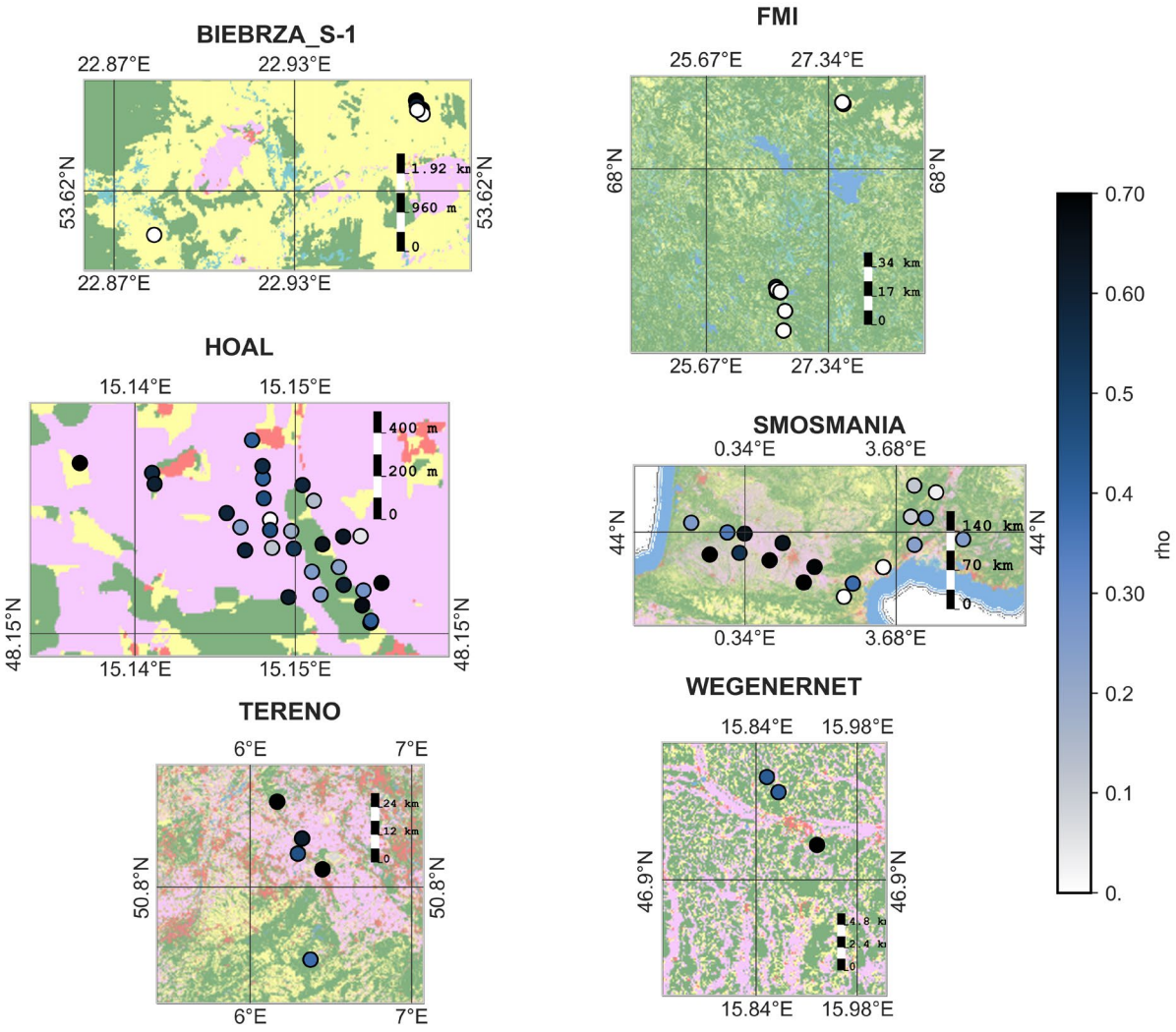


Figure 12: Pearson correlation (rho) for SWI1km with T-value=100 over selected ISMN networks (30-80 cm depth). See the land cover classes in Table 4.

© C-GLOPS Lot1 consortium

5.1.2 Summary over Networks: RMSD Analyses

In the following, the RMSD results are shown in the same way as in the above section on Pearson rho values. For the sake of clarity, and due to the high similarity with Pearson rho results, the spatial distribution of RMSD values within selected networks is omitted.

Not surprisingly when comparing Figure 4, Figure 5, Figure 7, Figure 8, Figure 10, Figure 11 with Figure 13 to Figure 18, the networks with small RMSD values generally correspond with networks with high correlations, and again, the differences between the soil depth layers are relatively small. One thing to notice is the low median RMSD at the BIEBRZA_S-1 network in the deep soil (Figure 17), which is much lower than in the near and sub-surface. Multiple stations from BIEBRZA_S-1 are located in wetlands where hydrological dynamics differ significantly from non-wetland areas. Another point is the larger spread of RMSD values over networks with in-situ stations located over multiple land covers (e.g. BIEBRZA-S1, HOAL), and over networks with stations scattered across a large territory (e.g. GROW, SMOSMANIA (deep soil)). At the near-surface (Figure 14) and sub-surface (Figure 16) layers, the RMSD values remain consistent in 2024 compared to the reference period. Concerning the deep soil layer (Figure 18), the differences of RMSD value between 2024 and the reference period show bigger changes: whereas FMI and NVE have higher RMSD, STEMS, and TERENO show lower values. RMSD for XMS-CAT stays the same. The larger changes of RMSD can be explained by lower number of stations in combination with the nature of the SWI approach and the uncertainty stemming from deriving soil moisture at deeper depth from satellite data. The RMSD values at the deep soil remain satisfactory regarding the challenging nature of modelling infiltration processes.

5.1.2.1 Near-surface Soil Layer (T-Value = 2)

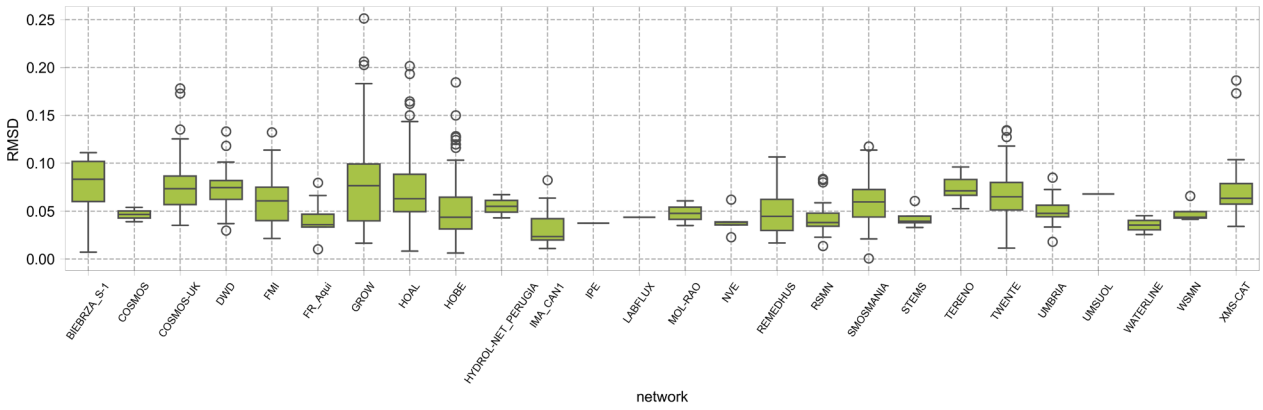


Figure 13: Boxplot summarizing RMSD values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) for the period between 2015 and 2024, per network.

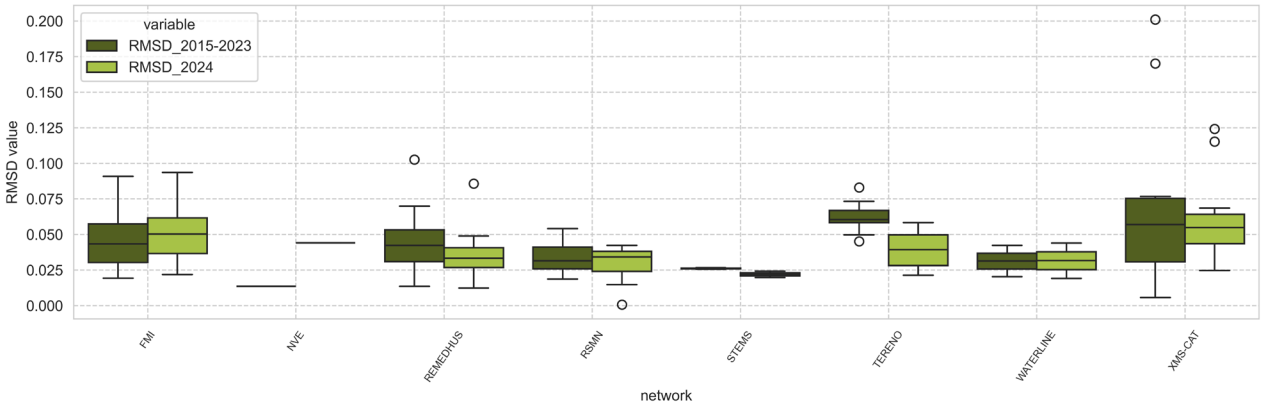


Figure 14: Box plot summarizing RMSD values calculated for SWI1km with T-value=2 and all relevant ISMN stations (0-10 cm depth) per network, based on data from 2015-2023 (dark green) and 2024 (light green). Only networks with data covering 2024 are included.

5.1.2.2 Sub-surface Soil Layer (T-Value = 10)

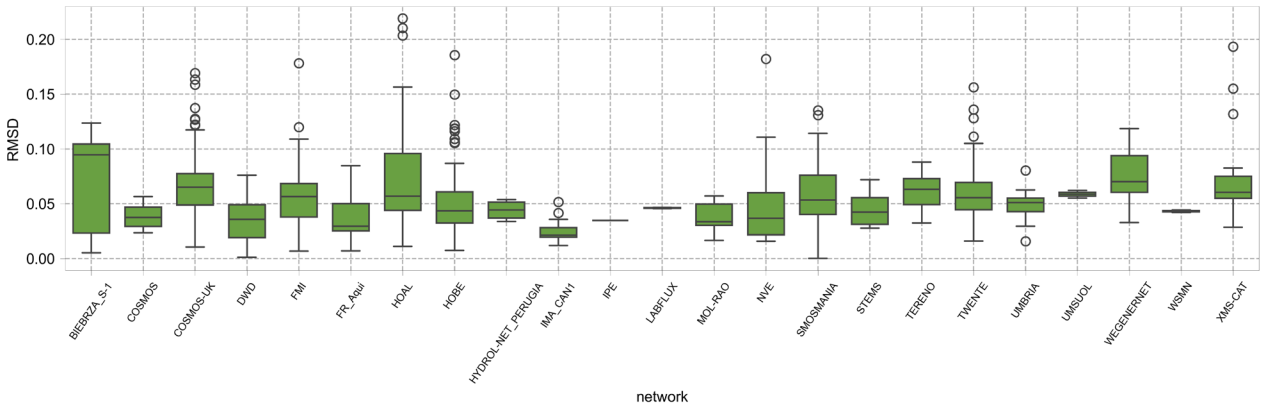


Figure 15: Boxplot summarizing RMSD values calculated for SWI1km with T-value=10 and all relevant ISMN stations (10-30 cm depth) for the period between 2015 and 2024, per network.

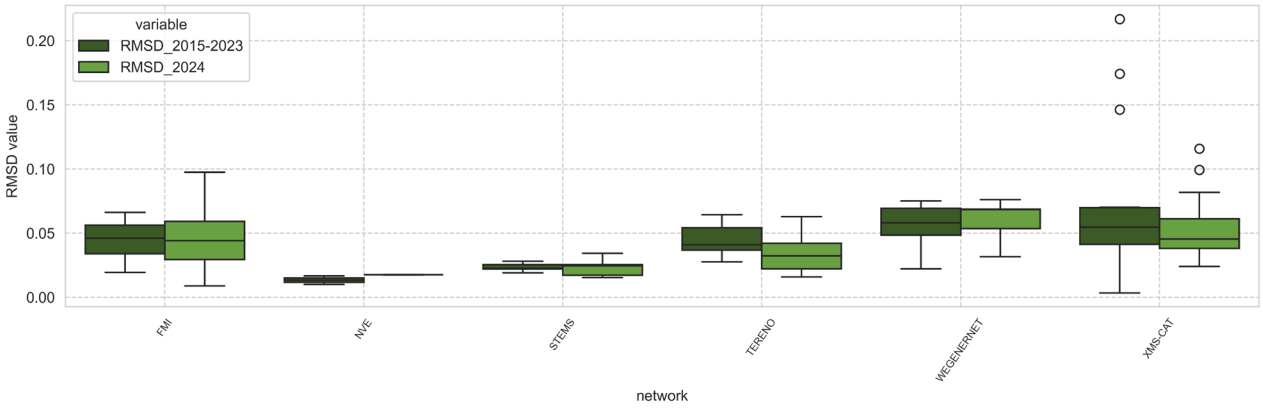


Figure 16: Box plot summarizing RMSD values calculated for SWI1km with T-value=10, and all suitable ISMN stations (10-30 cm depth) per network, based on data from 2015-2023 (dark green) and 2024 (light green). Only networks with data covering 2024 are included.

5.1.2.3 Deep Soil Layer (T-Value = 100)

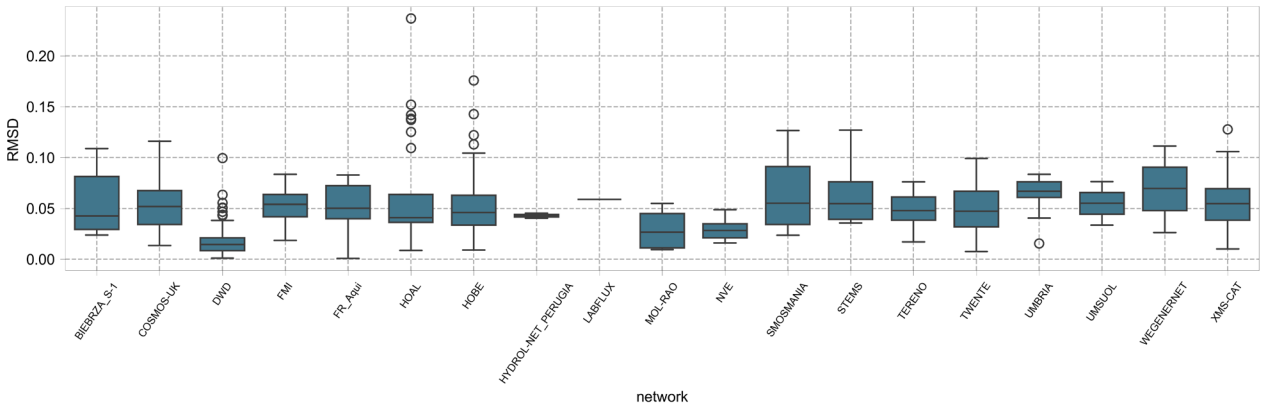


Figure 17: Boxplot summarizing RMSD values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) for the period between 2015 and 2024, per network.

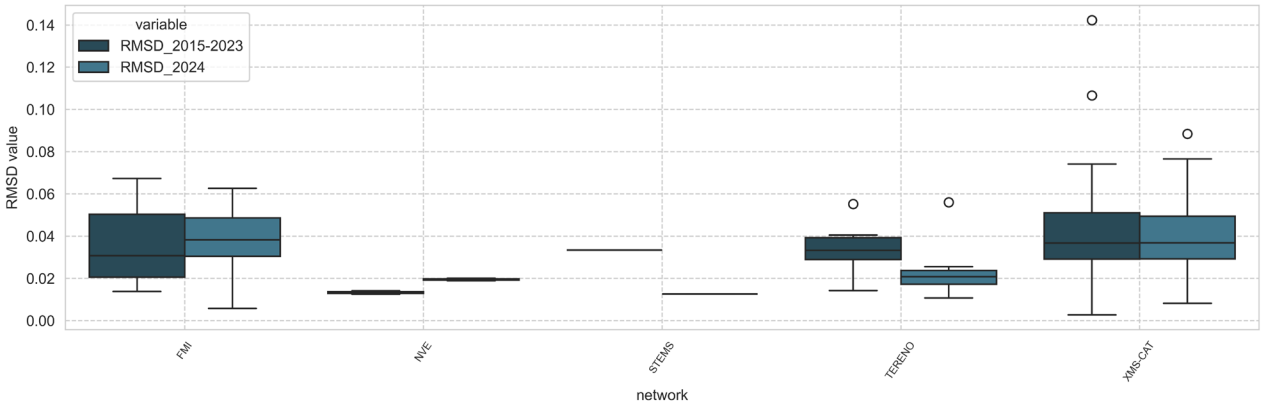


Figure 18: Box plot summarizing RMSD values calculated for SWI1km with T-value=100, and all suitable ISMN stations (30-80 cm depth) per network, based on data from 2015-2023 (dark blue) and 2024 (light blue). Only networks with data covering 2024 are included.

5.2 SELECTED TIME SERIES ANALYSES

This section gathers exemplary time series comparisons in order to illustrate the general temporal dynamics of satellite SWI1km and in-situ SM observations over multiple locations, land cover types, and depth levels. The availability of the stations can vary from some months per year to almost full coverage, and only few stations feature data for the full evaluation period thus affecting some performance metrics in individual networks.

5.2.1 Near-Surface Soil Layer (T-Value = 2)

The strongest dynamics can be observed in SWI1km satellite data and in-situ data in the near-surface layer naturally. When browsing through Figure 19 to Figure 28, one can generally see a very good agreement between the SWI1km and in-situ time series. The general temporal behaviour is depicted well, although the SWI1km time series show more short-term dynamics. From those time series, it is apparent the SWI1km product is especially suitable to reproduce soil moisture dynamics over relatively simple vegetation in temperate (e.g. RSMN, TERENO, COSMOS-UK) to semi-arid climate (e.g. REMEDHUS), as well as humid continental climates (e.g. XMS-CAT). In such environments, the radar signal is not dominated by complex scattering mechanisms due to vegetation geometry (e.g. dense forests) or masked due to freezing temperatures. We note that the XMS-CAT network spans a relatively large area with different climate classifications, with many stations placed in semi-arid regions. For this report, however, we chose to plot the station Serra De Costa Ampla (Figure 23) located within a region of humid continental climate.

In certain locations, such as the FMI network in northern Finland (Figure 19), the agreement is lower. On the one hand, this is because of large data gaps due to frozen soil and, on the other hand, due to land cover types over which the performance of SWI1km is known to be less reliable (e.g., temporarily frozen/melt-flooded surfaces and dense forests). In the time series plots of e.g. TERENO station Gevenich (Figure 20), REMEDHUS station La Cruz De Elias (Figure 25) and RSMN stations Iasi and Satu Mare (Figure 26, Figure 27), the known issue of a seasonal bias can be seen: due to an insufficient correction of vegetation effects in the ASCAT and Sentinel-1 retrieval algorithms, under- and overestimation occurs in spring, summer and fall (Hahn et al., 2017; Pfeil et al., 2018; Bauer-Marschallinger et al., 2018). This behaviour is often observed over cropland and grassland regions. However, as can be seen in Figure 22 (Wildernath station from TERENO), biases can be observed over deciduous broadleaf forest, a land cover class over which usually very distinct, complex radar scattering mechanisms occur, complicating the soil moisture estimation from backscatter (Pfeil et al.,

2020, Vreugdenhil et al., 2020). High agreements are observed over semi-arid areas as shown over REMEDHUS station Carretoro (Figure 24), here, soil moisture levels almost never exceed 0.15 m^3/m^3 and is mainly driven by precipitation events. Change in in-situ data behaviour can also impact the validation as shown in XMS-CAT station Serra de Costa Ampla (Figure 23). The timeseries displays a temporal gap and a change in its dynamics in 2024 (similar to the previous year, albeit smaller), which negatively affects the correlation. This is due to the absence of part of the growing season, resulting in reduced number of observations. Finally, comparing the SWI product against the new generation of in-situ SSM products, namely the COSMOS-UK network using cosmic-ray neutron sensors, yields promising results. These sensors have larger footprint (at the scale of hundreds of meters), reducing the spatial bias between satellite data and in-situ sensors. Figure 28 highlights the ability of SWI to capture SSM over a field in the UK with a correlation of 0.79.

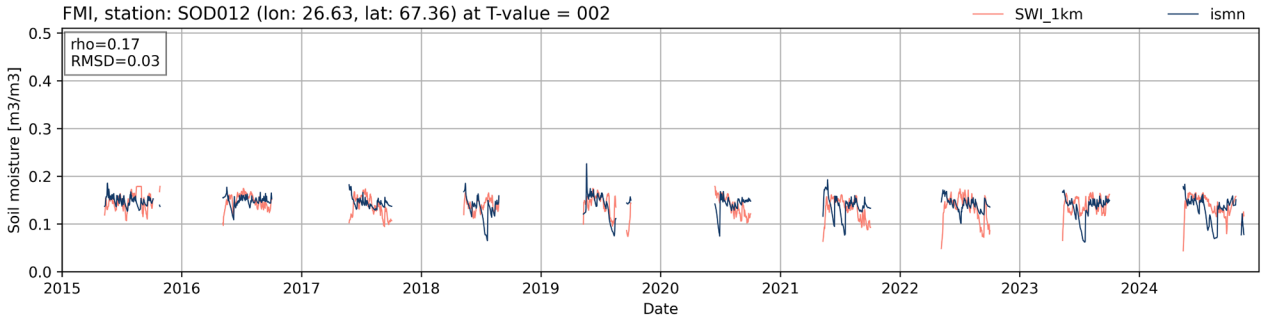


Figure 19: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at FMI station SOD012 at depth = 0.10m (tree cover – needle leaved). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left.

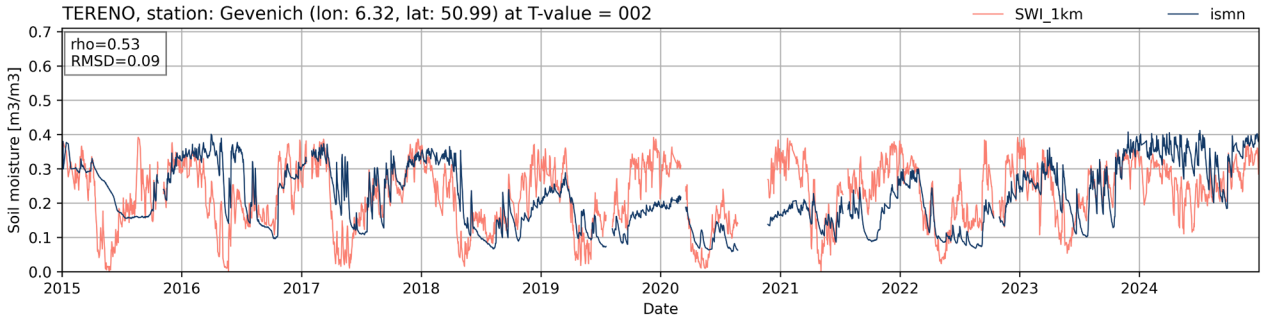


Figure 20: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at TERENO station Gevenich at depth = 0.05m (cropland - rainfed)

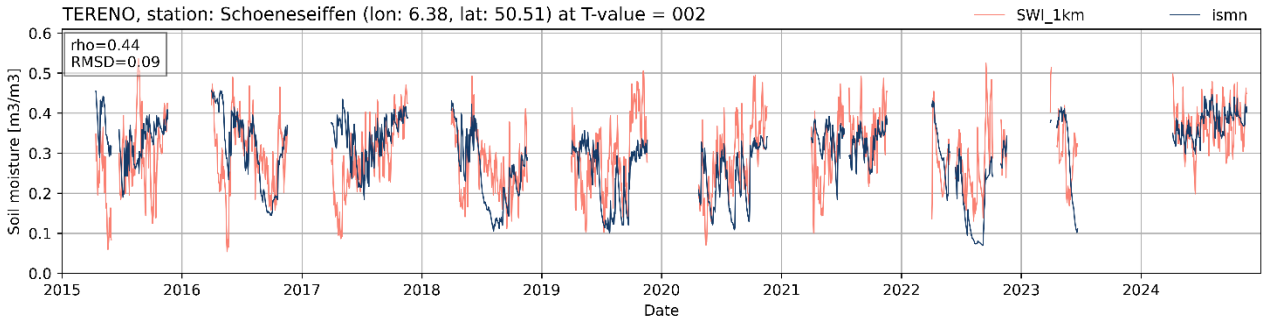


Figure 21: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at TERENO station Schoeneseiffen at depth = 0.20m (grassland).

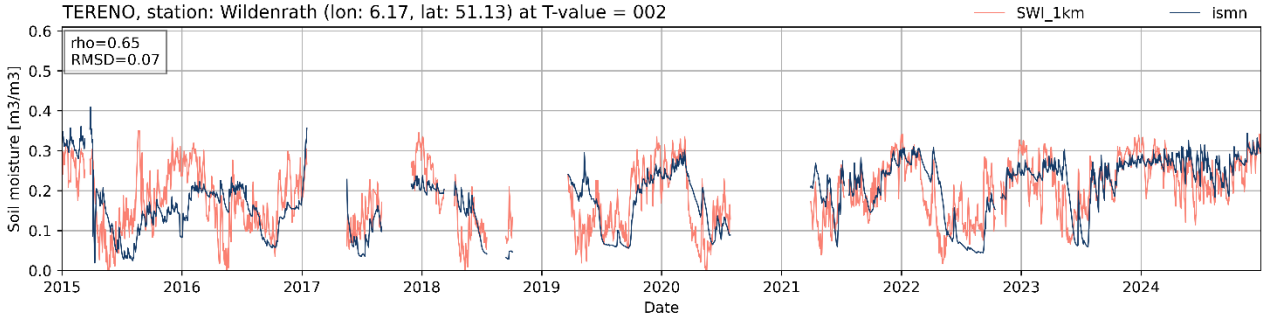


Figure 22: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at TERENO station Wildernath at depth = 0.05m (Forest – broad leaved).

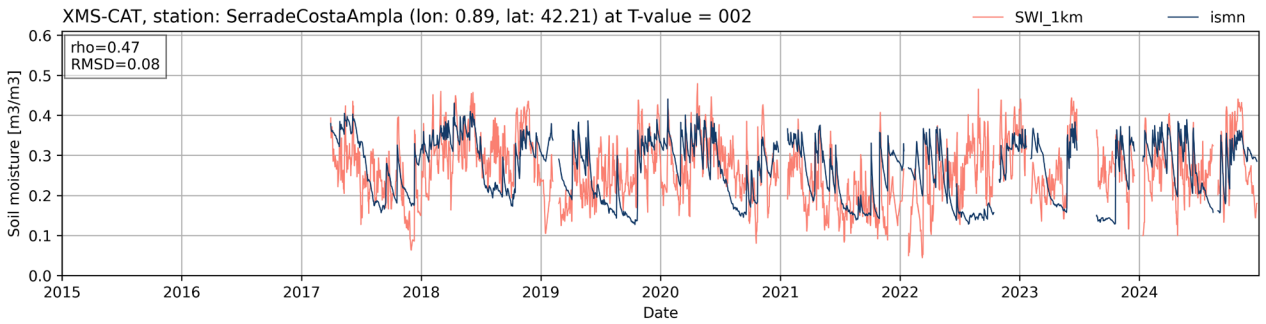


Figure 23: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at XMS-CAT station Serra De Costa Ampla at depth = 0.05m (cropland - rainfed).

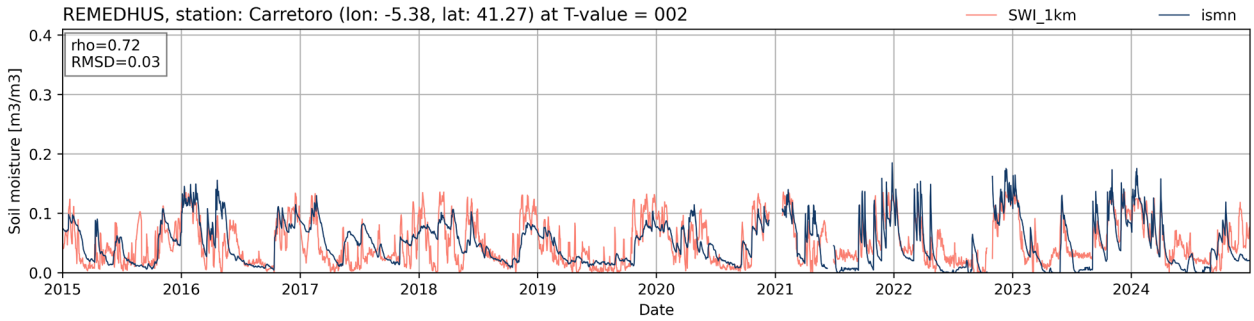


Figure 24: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at REMEDHUS station Carretoro at depth = [0-0.05]m (cropland - rainfed).

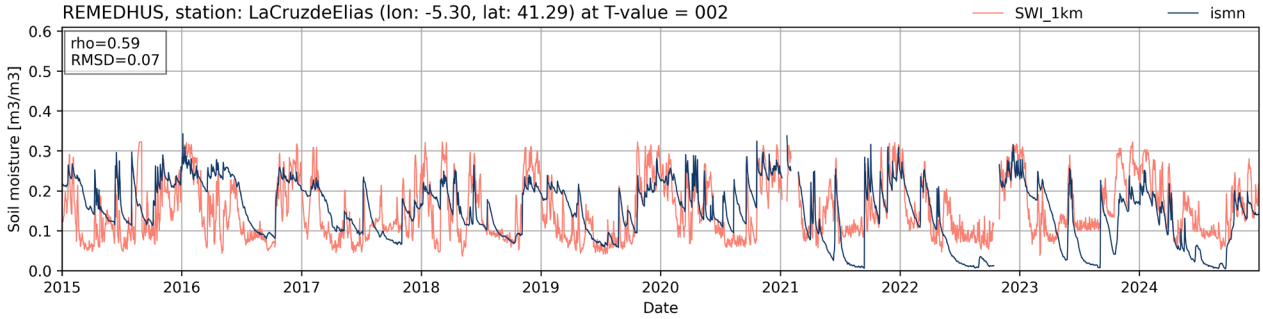


Figure 25: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at REMEDHUS station La Cruz De Elias at depth = [0-0.05]m (cropland - rainfed).

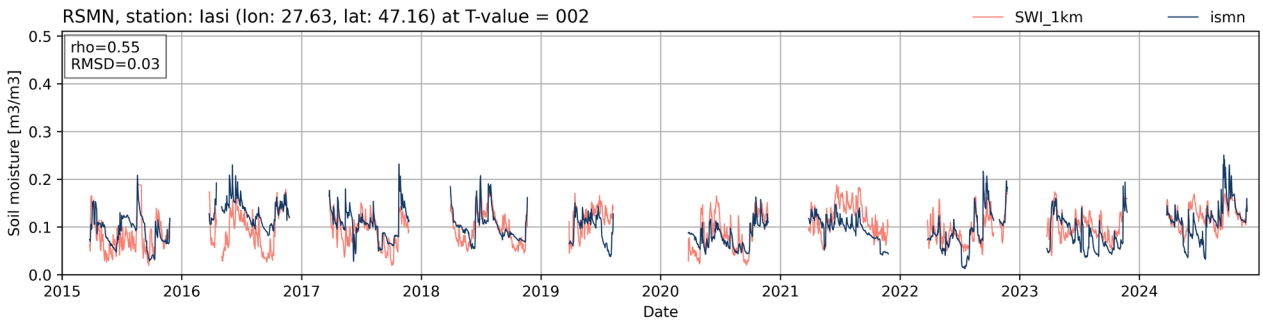


Figure 26: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at RSMN station Iasi at depth = [0-0.05]m (cropland - rainfed).

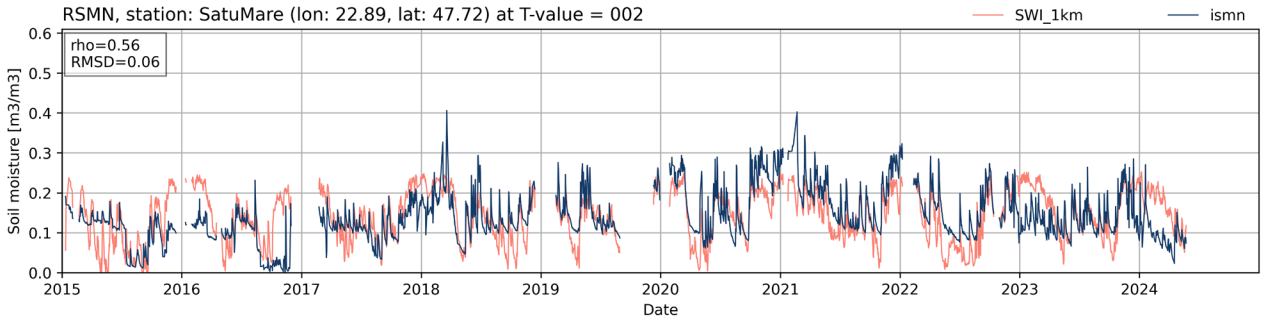


Figure 27: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at RMSN station Satu Mare at depth = [0-0.05]m (cropland - rainfed).

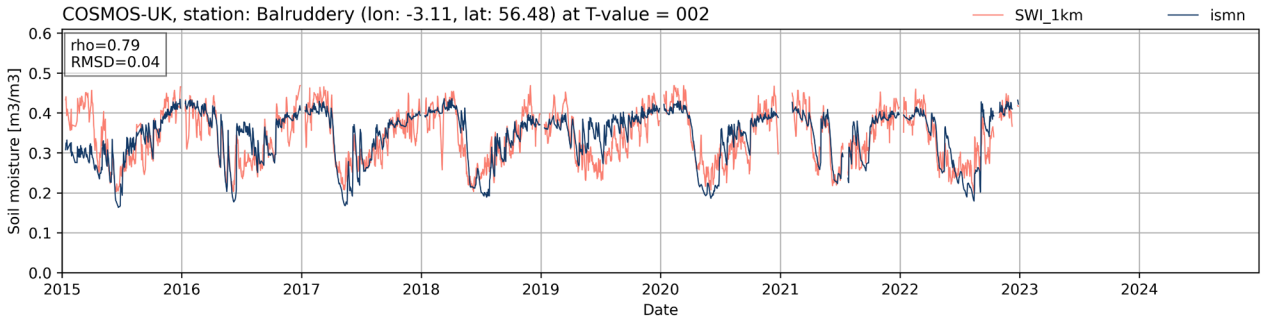


Figure 28: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=2; red) at COSMOS-UK station Balruderry at depth = 0.10m (cropland - rainfed).

5.2.2 Sub-Surface Soil Layer (T-Value = 10)

In the sub-surface layer, soil moisture shows less dynamics. In this layer, the general soil moisture signals agree again well for many stations, but the peaks and lows are not reproduced as accurately as in the near-surface layer. Considering that the SWI infiltration modelling can be seen only as an approximation of the local percolation rate, a perfect fit between T-value and soil layer depth cannot be assumed, and hence the inferior ability to reproduce peaks and lows could be explained.

As shown in the overall summary of the analysis (section 5.1), a good agreement is found over many networks. The average correlation of 0.57 (listed in Table 3) is higher than for the near-surface and the deep soil layer. Figure 29, Figure 31, and Figure 32 show as examples the very good agreements at STEMS station CAN-GC-T (rho=81, RMSD=0.03 m^3/m^3), TERENO stations

Wildenrath (rho=0.72, RMSD=0.05 m^3/m^3) and Selhausen (rho=0.78, RMSD=0.03 m^3/m^3), with the first showing a remarkable agreement, considering the known scale gap between point-shaped in-situ and the comparatively coarse satellite data. At some stations, however, there are seasonal biases again and signal delays are observed. For instance, at the XMS-CAT station Serra De Costa Ampla (Figure 33) and slightly for both TERENO's stations (Figure 31, Figure 32), the low soil moisture values are delayed in the in-situ data, suggesting lower than expected infiltration rate. The observed seasonal biases are generally negative (underestimated) in spring, and positive (overestimated) in summer. However, the magnitude of the biases varies throughout the year.

Figure 30 shows an example of the Finnish FMI station SOD101, where the comparison between satellite and in-situ data is only available over a few months, due to frozen soil during the rest of the year. The available in-situ time series show mostly a drying trend over the entire season, only interrupted by a few rainfall events. In SWI1km, this trend is not depicted well, and the time series shows more dynamics than the in-situ data. As mentioned before, this is most likely explained by the land cover in the region (needle leaf forests, close to inundated surfaces affecting the signal of coarse ASCAT observations (see Figure 9, for instance)).

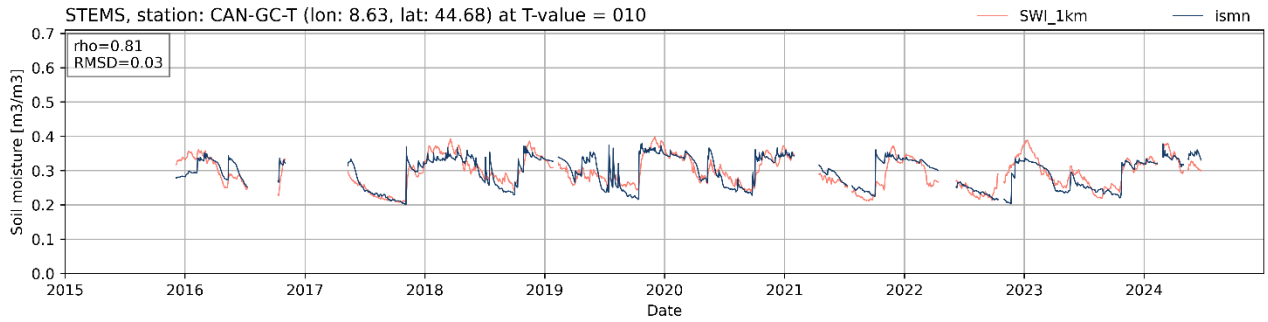


Figure 29: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at STEM station CAN-GC-T at depth = 0.30m (cropland - rainfed). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left.

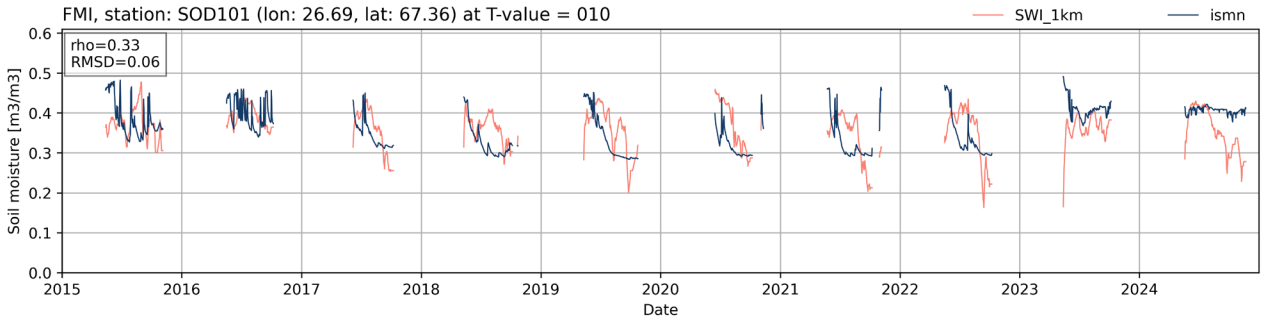


Figure 30: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at FMI station SOD101 at depth = 0.20m (tree cover – needle leaved).

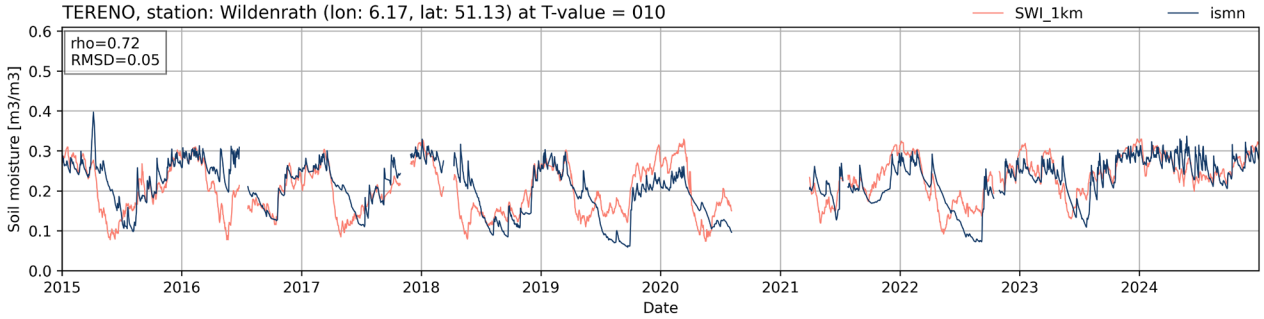


Figure 31: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at TERENO station Wildenrath at depth = 0.20m (Cropland - rainfed).

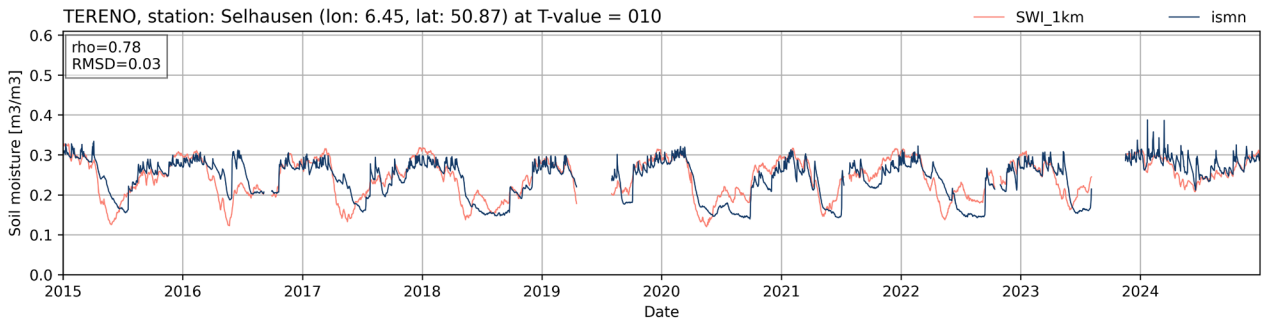


Figure 32: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at TERENO station Selhausen at depth = 0.20m (cropland - rainfed).

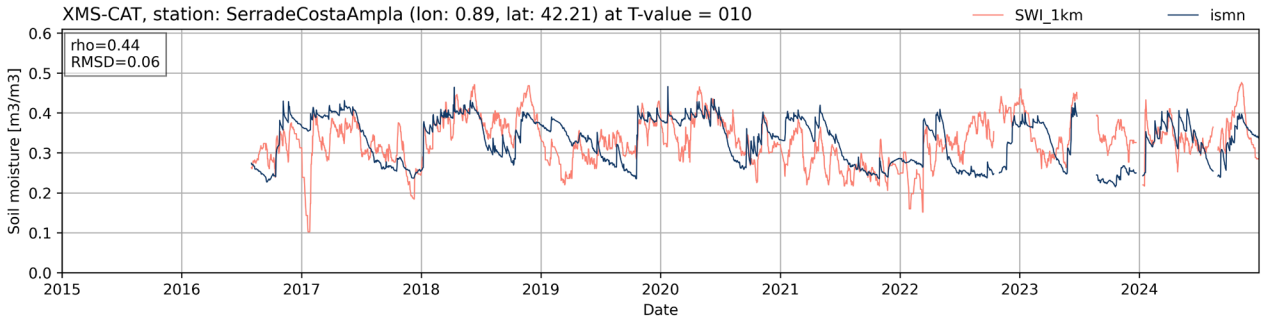


Figure 33: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=10; red) at XMS-CAT station SerraDeCostaAmpla at depth = 0.20m (cropland - rainfed).

5.2.3 Deep Soil Layer (T-Value = 100)

For the deep soil layer, fewer networks and stations provide measurements (which is a rather common observation for in-situ data). Both in-situ and SWI1km feature much smoother time series, fulfilling expectations on moisture dynamics in deeper layers. For many stations, the agreement is more than satisfying considering the expected spatial representativeness, the uncertainty of the relationship between T-value and depth, and the completely different measurement technology of the compared systems (e. g. Figure 35, Figure 36, Figure 38).

Some peaks are missed by the SWI1km data as a direct result of the heavy temporal filtering during the SWI calculation with a T-value=100, revealing the per-definition incapability of the SWI approach to capture strong rainfall percolation that reach deeper soil layers in a short time frame. For small T-values, this is not the case, and hence the SWI can potentially record strong dynamics in shallower soil layers if detected by the input SSM data. As an example, the station named “As” from the network NVE is shown (Figure 34). At this site, long-term dynamics are well captured, but short-term dynamics are almost entirely missed.

An example which shows that T-values, and thus the associated temporal smoothing, need to be carefully chosen is the TERENO station Gevenich (Figure 39). There, the soil has a high sand content (over 80%), and water percolates to deeper layers easily. Thus, surface soil moisture levels drop quickly, as can be seen in near-surface analysis in Figure 20. In the deeper soil however, high short-term dynamics occur, which are missed by SWI1km with a T-value of 100. The long-term dynamics in SWI1km and the in-situ time series do, however, agree very well, while time lag corresponding to

the quicker percolation rate is apparent. A similar example of missed short-term dynamics in deep soil layer can also be observed in XMS-CAT station Batlliu de Sort (Figure 37).

As aforementioned, the relationship between T-value and depth is not straightforward and varies between sites. For example, in some cases, the T-value 100 is not optimal for the selected deep soil layer of the ISMN data, which manifests itself as a time lag between SWI1km and in-situ time series. This is illustrated here for the STEMS network, station CAN-GC-T. Here, a positive time lag is observed, meaning that the dynamics in SWI1km are shifted and occur later than in the in-situ time series (Figure 35). This is expressed by a relatively low correlation value, compared to the time-series at the same location for shallower depths (compare to Figure 29). Depending on the local soil composition and layering, the observed time-lag is not consistent over the different stations. Similar lags can be seen in other stations, such as XMS-CAT station Serra de Costa Ampla (Figure 36).

Seasonal biases are exemplified by the TERENO station Schoeneseiffen (Figure 38). In the deep soil layer, the seasonal effect is less clearly distinguishable from biases than in the near surface layer (Figure 21). There, the temporal smoothing of the SWI induces on default a (desired) time-lag that can lead to a too strong temporal shift in case of a mismatch between the actual soil depth and the SWI's T-Value.

At T-value=100, the FMI station shows even negative correlation (Figure 40). As explained in section 5.1.1, the quality of the validation is hampered by the challenging masking of frozen soil and dense vegetation land cover.

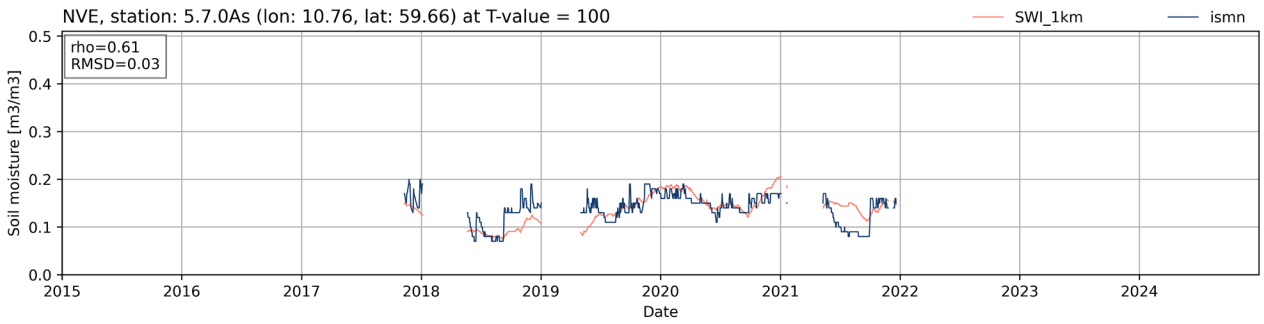


Figure 34: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at NVE station As at depth = 0.50m (cropland - rainfed). The Pearson correlation coefficient (rho) and RMSD value are given in the upper left.

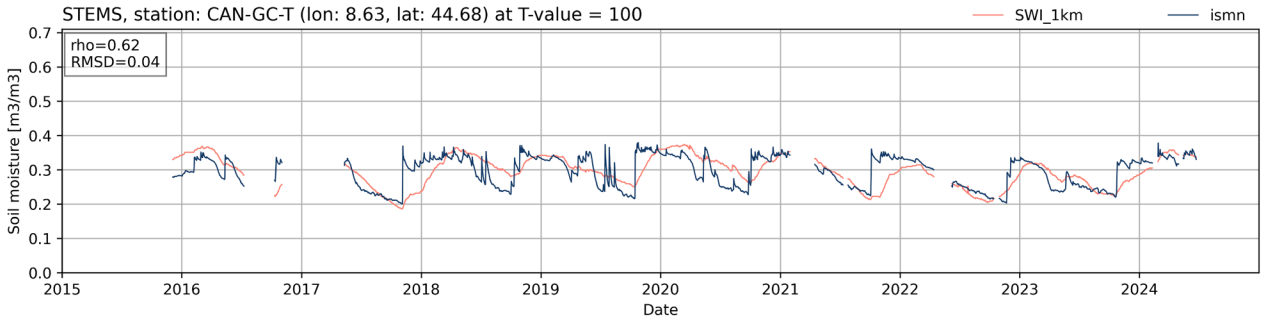


Figure 35: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at STEMS station CAN-GC-T at depth = 0.30m (cropland - rainfed).

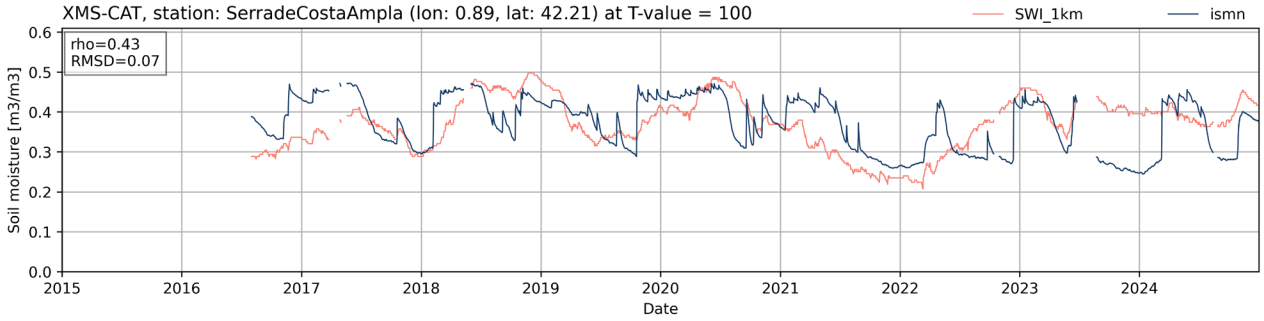


Figure 36: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at XMS-CAT station Serra de Costa Ampla at depth = 0.50m (cropland - rainfed).

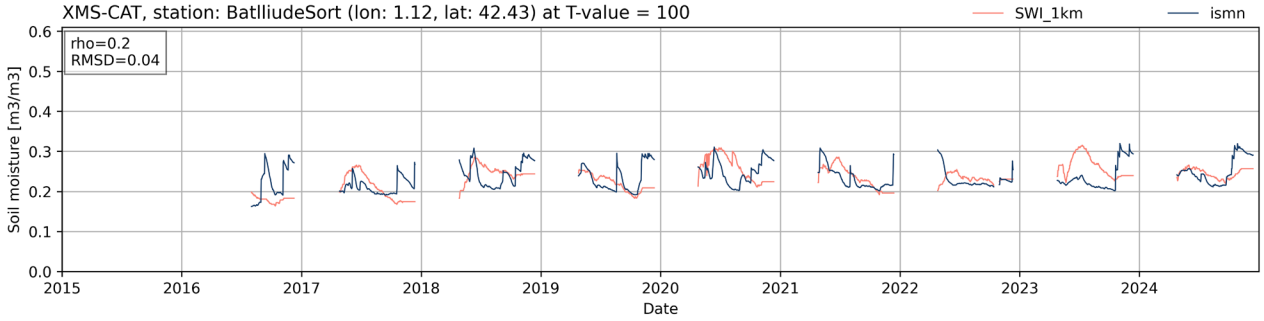


Figure 37: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at XMS-CAT station Batlliu de Sort at depth = 0.50m (grassland).

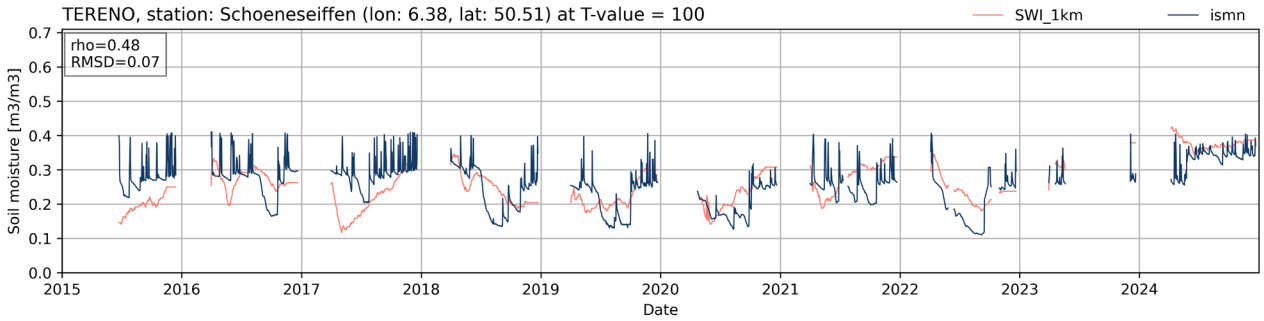


Figure 38: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at TERENO station Schoeneseiffen at depth = 0.70m (grassland).

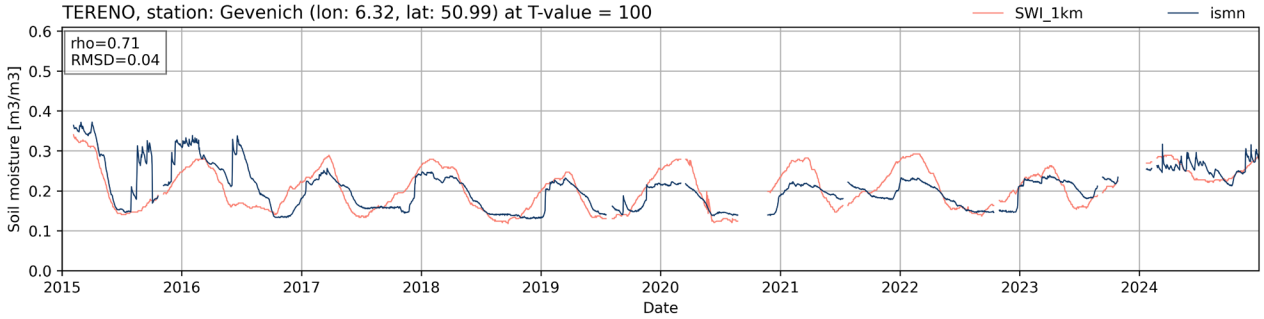


Figure 39: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at TERENO station Gevenich at depth = 0.30m (cropland).

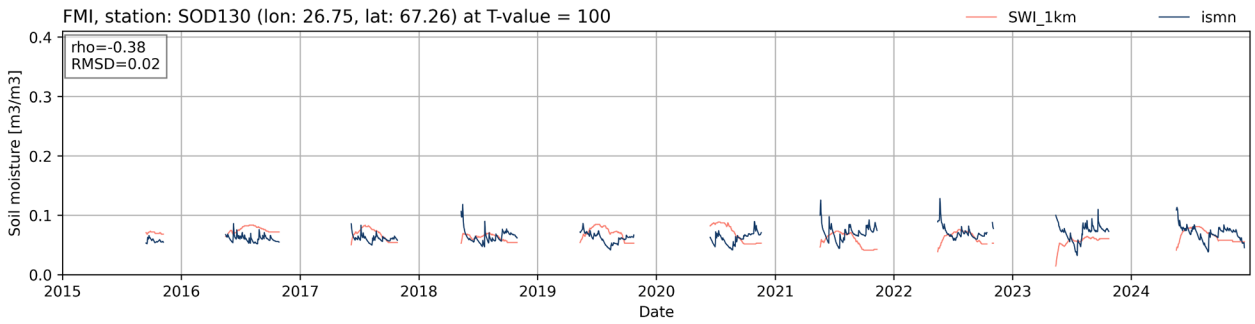


Figure 40: Temporal evolution of in-situ SM (blue) and SWI1km (T-value=100; red) at FMI station SOD130 at depth = 0.40m (cropland).

6 DISCUSSIONS AND CONCLUSIONS

The CLMS SWI1km product combines radar soil moisture estimates from Metop ASCAT and Sentinel-1 and aims to provide SWI estimates with a high temporal and spatial resolution.

This section first highlights the in-depth study of Bauer-Marschallinger et al. (2018b) over Italy in the period 2015-2017, and then draws the conclusions from the here presented (larger) quality evaluation over Europe in the period 2015-2024.

6.1 SUMMARY FROM DEDICATED PEER-REVIEW ARTICLE

The near-real-time-capable SCATSAR-SWI algorithm (that is used in CLMS for the SWI1km production) is applied on a fused 3-year ASCAT-Sentinel-1-SSM data cube over Italy, yielding a consistent set of model parameters, unperturbed by coverage discontinuities. An evaluation of the obtained SWI1km dataset, involving a 1 km Soil Water Balance Model (SWBM) over Umbria (Italy), yields comprehensively high agreement with the reference data (median R = 0.61 vs. in situ; 0.71 vs. model; 0.83 vs. ASCAT SSM). While the Sentinel-1 signal is attenuated to some extent, ASCAT's signal dynamics are fully transferred to the SWI1km and benefit from the Sentinel-1 parametrisation. Using the SM2RAIN approach to model rainfall from satellite SM data, the SWI1km shows excellent capability to reproduce 5 day-accumulated rainfall over Italy, with R=0.89 against observed rainfall. All details on this analysis can be found in Bauer-Marschallinger et al. (2018b).

6.2 QUALITY ASSESSMENT OVER EUROPE 2015-2024

In this report, the statistical evaluation of the SWI1km product was conducted for the years 2015-2024, with a focus on 2024, using ISMN stations distributed all over Europe. At the ISMN networks with fully updated data in 2024 (FMI, NVE, REMEDHUS, RSMN, STEMS, TERENO, WATERLINE, WEGENERNET, XMS-CAT), the statistical time series comparisons with in-situ soil moisture stations showed an overall satisfying result, with an overall high performance that is similar or even improved compared to the ASCAT-alone based product SWIV3 [CGLOPS1_VR2024_SWIV3], and a clearly improved performance compared to the Sentinel-1-alone based product [CGLOPS1_VR2024_SSM1km].

For the near-surface and sub-surface layers (0.0-0.1m and 0.1-0.3m, T-value=2 and 10), the average Pearson coefficient (rho) value is 0.54 and 0.57, with average RMSD values of 0.07 and 0.06 m^3/m^3

respectively. Also, due to the scale gap between in-situ point measurements and the satellite product's 1km sampling and the consequent representativeness errors, deviations are to be expected.

The scores for deeper soil layers (down to 0.80 m) are also considered acceptable with an average Pearson coefficient (rho) value of 0.51 and an average RMSD of 0.05 m^3/m^3. The comparison of data from the year 2024 versus data from 2015-2023 maintained a consistent data quality for nearly all networks and soil layers, also confirming what can be expected from the increasingly improved quality of the Sentinel-1 input data in the recent years.

A seasonal bias appears at some stations, showing a partly large under- or overestimation of soil moisture during the growing season. This issue seems to be inherited from the input ASCAT SSM and Sentinel-1 SSM datasets, which both suffer from a misinterpretation of vegetation density change and vitality dynamics in the radar backscatter modelling prior to SSM estimation (Hahn et al., 2017; Pfeil et al., 2018; Bauer-Marschallinger et al., 2018).

In cold climates, more specifically in northern Finland, the agreement with ground measurements was found to be rather low. This is most likely caused by the evergreen needle leaf forest and partly inundated land cover dominating the region, as well as frozen surface conditions. However, as part of the upcoming updates of the currently ongoing CLMS evolution activities, a masking process based on meteorological data is developed to mask out SSM retrievals irritated by frozen surface conditions. This could improve the agreement of the next product version in northern regions with colder climates.

The overall performance of the SWI1km product is consistent with the previous years' scientific evaluation experiments. No inconsistencies in the SWI were detected in the current scientific quality monitoring exercise. The comparison between the reference period (2015-2023) and the current validation period (2024) demonstrated the product consistency and suggests that SM conditions in 2024 are well represented by the algorithm.

The loss of Sentinel-1B in late 2021 has been acknowledged, while Sentinel-1A is stably observing, and Sentinel-1C was launched successfully at the end of 2024. In the data for 2022-2024, it could be confirmed that the SWI-approach is able to cope with the inconsistencies in the spatio-temporal coverage of its SAR component. The SWI1km signal quality is only weakly affected, with no degradation of the scores. Most notably, a few cases of missed short-term dynamics after rainfall events are apparent.

7 RECOMMENDATIONS AND PERSPECTIVES

As the product quality is stable or even improving in the last year, we recommend a continuation of using the SWI1km product. In general, the products appear to have higher fidelity in the shallower soil layers, concerning the timing of peaks and lows. The masking for frozen or snow-covered soil is currently still estimated at the 12.5km sampling and thus might lead to not optimal local flagging and users are advised to consult additional data when analysing data from cold seasons. It should be moreover noted that, depending on the vegetation density, biases can appear over densely vegetated areas (e.g. forests) or agricultural areas.

Future product enhancements are addressing the know issues (vegetation biases, need for masking of frozen soil conditions, topography and observation geometry effects) and will be implemented through currently ongoing developments within the CLMS evolution activities, leading to an enhanced Sentinel-1 SAR backscatter preprocessing (including Sentinel-1C observations) and SSM modelling, which includes dynamic vegetation correction, an enhanced masking for dense vegetation, subsurface-scattering and water bodies. Furthermore, the new upgrades of the EUMETSAT H SAF SSM product is based on the ASCAT Level 1b full-resolution backscatter product, profiting from a higher spatial resolution of this input, an improved vegetation characterisation, masking of ill-posed subsurface scattering areas, and a dynamic parameterization of the dry- and wet-references to correct for slow-acting trends coming from long-term land cover changes. In addition to these new H SAF upgrades, the Surface State Flag will undergo a refinement, taking up the newly available ECWMF NRT forecast data for frozen surfaces and snow cover. The CLMS users can therefore expect improved quality and reliability of the future SWI1km product version 2.

8 REFERENCES

Azimi, S., Dariane, A. B., Modanesi, S., Bauer-Marschallinger, B., Bindlish, R., Wagner, W., & Massari, C. (2020). Assimilation of Sentinel 1 and SMAP–based satellite soil moisture retrievals into SWAT hydrological model: the impact of satellite revisit time and product spatial resolution on flood simulations in small basins. Journal of Hydrology, 581, 124367.

Bauer-Marschallinger, B., Naeimi, V., Cao, S., Paulik, C., Schaufler, S., Stachl, T., Modanesi, S., Ciabatta, L., Massari, C., Brocca, L. & Wagner, W. (2018). Toward Global Soil Moisture Monitoring With Sentinel-1: Harnessing Assets and Overcoming Obstacles. IEEE Transactions on Geoscience and Remote Sensing, 57, 520-539. doi:10.1109/TGRS.2018.2858004

Bauer-Marschallinger, B., Paulik, C., Mistelbauer, T., Hochstöger, S., Modanesi, S., Ciabatta, L., Massari, C., Brocca, L. & Wagner, W. (2018b). Soil Moisture from Fusion of Scatterometer and SAR: Closing the Scale Gap with Temporal Filtering. Remote Sensing, 10(7), 1030. doi:10.3390/rs10071030

Brocca, L., Hasenauer, S., Lacava, T., Melone, F., Moramarco, T., Wagner, W., Dorigo, W., Matgen, P., Martínez-Fernández, J., Llorens, P., Latron, J., Martin, C. & Bittelli, M. (2011). Soil moisture estimation through ASCAT and AMSR-E sensors: An intercomparison and validation study across Europe. Remote Sensing of Environment, 115, 3390-3408.

Crow, W. T., Berg, A. A., Cosh, M. H., Loew, A., Mohanty, B. P., Panciera, R., ... & Walker, J. P. (2012). Upscaling sparse ground-based soil moisture observations for the validation of coarse-resolution satellite soil moisture products. Reviews of Geophysics, 50(2).

De Lannoy, G. J., Pauwels, V. R., Houser, P. R., Gish, T., & Verhoest, N. E. (2007). Representativeness of point soil moisture observations, upscaling and assimilation. IAHS PUBLICATION, 313, 249.

Dorigo, W.A., Wagner, W., Hohensinn, R., Hahn, S., Paulik, C., Xaver, A., Gruber, A., Drusch, M., Mecklenburg, S., van Oevelen, P., Robock, A., and Jackson, T., Jackson, T. (2011). The International Soil Moisture Network: A data hosting facility for global in situ soil moisture measurements, Hydrology and Earth System Sciences 15 (5), pp. 1675-1698

Dorigo, W.A., Xaver, A., Vreugdenhil, M., Gruber, A., Hegyiová, A., Sanchis-Dufau, A.D., Zamojski, D., Cordes, C., Wagner, W., & Drusch, M. (2013). Global Automated Quality Control of In Situ Soil Moisture Data from the International Soil Moisture Network. Vadose Zone Journal, 12

Dorigo, W., Himmelbauer, I., Aberer, D., Schremmer, L., Petrakovic, I., Zappa, L., ... & Sabia, R. (2021). The International Soil Moisture Network: serving Earth system science for over a decade. Hydrology and Earth System Sciences Discussions, 1-83.

Gruber, A., Dorigo, W. A., Zwieback, S., Xaver, A., & Wagner, W. (2013). Characterizing coarse-scale representativeness of in situ soil moisture measurements from the International Soil Moisture Network. Vadose Zone Journal, 12(2).

Hahn, S., Reimer, C., Vreugdenhil, M., Melzer, T., & Wagner, W. (2017). Dynamic characterization of the incidence angle dependency of backscatter using Metop ASCAT. IEEE Journal of Selected Topics in Applied Earth Observation and Remote Sensing, 10, 2348-2359

Lima, F. T., & Souza, V. M. (2023). A large comparison of normalization methods on time series. Big Data Research, 34, 100407.

Montzka, C., Cosh, M., Bayat, B., Al Bitar, A., Berg, A., Bindlish, R., ... & Nickeson, J. (2020). Good Practices for Satellite Derived Land Product Validation. Soil Moisture Product Validation Good Practices Protocol Version 1.0.

Paulik, C., Dorigo, W., Wagner, W., & Kidd, R. (2014). Validation of the ASCAT Soil Water Index using in situ data from the International Soil Moisture Network. International Journal of Applied Earth Observation and Geoinformation, 30, 1-8

Pfeil, I., Vreugdenhil, M., Hahn, S., Wagner, W., Strauss, P., & Blöschl, G. (2018). Improving the seasonal representation of ASCAT soil moisture and vegetation dynamics in a temperate climate. Remote Sensing, 10

Pfeil, I., Wagner, W., Forkel, M., Dorigo, W., & Vreugdenhil, M. (2020). Does ASCAT observe the spring reactivation in temperate deciduous broadleaf forests? Remote Sensing of Environment, 250, 112042.

Termite, L. F., Garinei, A., Marini, A., Marconi, M., & Biondi, L. (2019). Combining satellite data and Machine Learning techniques for irrigation Decision Support Systems. In 2019 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor) (pp. 291-296). IEEE.

Vreugdenhil, M., Navacchi, C., Bauer-Marschallinger, B., Hahn, S., Steele-Dunne, S., Pfeil, I., ... & Wagner, W. (2020). Sentinel-1 Cross Ratio and Vegetation Optical Depth: A Comparison over Europe. Remote Sensing, 12(20), 3404.